

Classification of Complex Semisimple Lie Algebras

Labix

February 26, 2026

Abstract

Potentially good books: Humphreys, Erdmann and Wildson

Contents

1	Weights and Weight Spaces	3
1.1	Weights	3
1.2	Weight Spaces	3
2	Lie Algebras over a Field	5
2.1	Lie Subalgebras of the Matrix Ring	5
2.2	Nilpotency and Engel's Theorem	6
2.3	Solubility and Lie's Theorem	9
2.4	Low Dimensional Lie Algebras	11
3	The Killing Form of a Lie Algebra	13
3.1	The Killing Form	13
3.2	Cartan's Two Criteria	14
4	Special Subalgebras of a Lie Algebra	18
4.1	Derivations of a Semisimple Lie Algebras over $\mathbb{C}$	18
4.2	Cartan Subalgebras	19
5	Representation Theory of Lie Algebras	21
5.1	Weyl's Theorem	21
5.2	Representations of Lie Algebras	21
5.3	The Case of $\mathfrak{sl}(2, \mathbb{C})$	21
6	Classification of Semisimple Lie Algebras over $\mathbb{C}$	25
6.1	Root Space Decomposition	25
6.2	Copies of $\mathfrak{sl}_2(\mathbb{C})$ as Sub-algebras	27
6.3	The Dimension of the Subspaces in Root Space Decompositions	29
6.4	Root Strings	31
6.5	The Set of Roots as a Root System	32
6.6	The Classification Theorem	32
6.7	Simple Lie Algebras and Irreducible Dynkin Diagrams	34

1 Weights and Weight Spaces

1.1 Weights

Definition 1.1.1 (Common Eigenvectors)

Let k be a field. Let V be a vector space over k . Let W be a vector subspace of $\text{End}_k(V)$. A common eigenvector of W is a vector $v \in V$ such that for all $T \in W$, v is an eigenvector of T .

Definition 1.1.2 (Weights)

Let k be a field. Let V be a vector space over k . Let W be a vector subspace of $\text{End}_k(V)$. Let $v \in V$ be an eigenvector of W . A weight of W for v is an assignment $\lambda : W \rightarrow k$ such that the eigenvalue of T corresponding to v is $\lambda(T)$.

Lemma 1.1.3

Let k be a field. Let V be a vector space over k . Let W be a vector subspace of $\text{End}_k(V)$. Let $\lambda : W \rightarrow k$ be a weight of W . Then $\lambda \in W^*$.

1.2 Weight Spaces

Definition 1.2.1 (Weight Spaces)

Let k be a field. Let V be a vector space over k . Let W be a vector subspace of $\text{End}_k(V)$. Let $\lambda \in W^*$ be a weight of W . Define the weight space of λ to be

$$V_\lambda = \{v \in V \mid T(v) = \lambda(T)v \text{ for all } T \in W\}$$

Lemma 1.2.2

Let k be a field. Let V be a vector space over k . Let W be a vector subspace of $\text{End}_k(V)$. Let $\lambda \in W^*$ be a weight of W . Then V_λ is a vector subspace of V .

Proposition 1.2.3

Let k be a field. Let V be a vector space over k . Let L be a Lie-subalgebra of $\mathfrak{gl}_k(V)$. Let I be an ideal of L . Then

$$V_0 = \{v \in V \mid T(v) = 0 \text{ for all } T \in I\}$$

is an L -invariant subspace of V .

Proof Let $y \in L$ and $m \in V_0$. We want to show that $y(m) \in V_0$. So let $T \in I$ be arbitrary. We compute that

$$T(y(m)) = T(y(m)) - y(T(m)) = [T, y](m)$$

since the bracket of I is the bracket of L , which is the bracket of $\mathfrak{gl}(V)$. Also, $[T, y](m) = 0$ since I is an ideal implies that $[T, y] \in I$ so that $m \in \ker([T, y])$ since $m \in V_0$. Thus V_0 is an invariant subspace of L . ■

Proposition 1.2.4

Let k be a field. Let V be a finite dimensional vector space over k . Let L be a Lie-subalgebra of $\mathfrak{gl}_k(V)$. Let I be an ideal of L . Let λ be a weight of I . Then V_λ is an L invariant subspace of V .

Proof Let $x \in L$ and $w \in V_\lambda$. We want to show that $x(w) \in V_\lambda$. So let $T \in I$. We compute that

$$T(x(w)) = [T, x](w) + x(T(w)) = \lambda([T, x])w + \lambda(T)x(w)$$

since $[T, x] \in I$. Therefore it suffices to show that $\lambda([T, x]) = 0$.

Consider the subspace

$$W = \langle x^k(w) \mid k \in \mathbb{N} \rangle \subseteq V$$

Since V is finite dimensional, there exists $k \in \mathbb{N}$ such that $\{w, x(w), \dots, x^k(w)\}$ is a basis for W . I claim that for any W is I -invariant. Let $T \in I$. We inductively show that $T(x^i(w)) \in W$. When $i = 0$, $T(w) = \lambda(T)(w) \in W$. Then we also have

$$T(x^i(w)) = [T, x](x^{i-1}(w)) + x(T(x^{i-1}(w))) = \lambda([T, x])x^{i-1}(w) + \lambda(T)x^i(w)$$

by applying the inductive hypothesis. Then $x^{i-1}(w), x^i(w) \in W$ implies that $T(x^i(w)) \in W$. Then the matrix of the restriction of $T \in I$ to W with respect to our basis of W is given by

$$T = \begin{pmatrix} \lambda(T) & * & * & \cdots & * \\ 0 & \lambda(T) & * & \cdots & * \\ 0 & 0 & \lambda(T) & \cdots & * \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \lambda(T) \end{pmatrix}$$

Since this is true for any $T \in I$, we can apply this to any $z = [T, x] \in [L, I] \subseteq I$ to get that $\text{tr}(z) = (k+1)\lambda(z)$. Now $\text{tr}(z) = \text{tr}([T, x]) = \text{tr}(Tx - xT) = 0$ so that $(k+1)\lambda(z) = 0$. Hence $\lambda([T, x]) = \lambda(z) = 0$ as required. ■

2 Lie Algebras over a Field

2.1 Lie Subalgebras of the Matrix Ring

Proposition 2.1.1

Let k be a field. Let V be a Lie algebra over k . Choose a basis $\{e_1, \dots, e_n\}$ of V . Then the map $\mathfrak{gl}_n(V) \rightarrow \mathfrak{gl}_n(k)$ defined by

$$T \mapsto \begin{pmatrix} T(e_1) & \cdots & T(e_n) \end{pmatrix}$$

is a Lie algebra isomorphism.

Lemma 2.1.2

Let V be a vector space over a field k . Let $T, S, R \in \mathfrak{gl}_k(V)$. Then

$$\mathrm{tr}([T, S] \circ R) = \mathrm{tr}(T \circ [S, R])$$

Definition 2.1.3 (Lie Sub-algebra of 0 trace)

Let k be a field. Let $n \in \mathbb{N} \setminus \{0\}$. Define

$$\mathfrak{sl}_n(k) = \{A \in \mathfrak{gl}_n(k) \mid \mathrm{tr}(A) = 0\}$$

Definition 2.1.4 (Lie Sub-algebra of Upper Triangular Matrices)

Let k be a field. Let $n \in \mathbb{N} \setminus \{0\}$. Define the Lie sub-algebra of upper triangular matrices to be

$$\mathfrak{b}_n(k) = \{A \in \mathfrak{gl}_n(k) \mid \mathrm{tr}(A) = 0\}$$

Definition 2.1.5 (Lie Sub-algebra of Strictly Upper Triangular Matrices)

Let k be a field. Let $n \in \mathbb{N} \setminus \{0\}$. Define the Lie sub-algebra of strictly upper triangular matrices to be

$$\mathfrak{u}_n(k) = \{A \in \mathfrak{gl}_n(k) \mid \mathrm{tr}(A) = 0\}$$

Definition 2.1.6 (Symplectic Lie Algebra)

Let k be a field. Let $n \in \mathbb{N}$. Define the symplectic Lie algebra to be

$$\mathfrak{sp}_{2n} = \left\{ \begin{pmatrix} A & B \\ C & -A^T \end{pmatrix} \in \mathfrak{gl}_{2n}(k) \mid A, B, C \in \mathfrak{gl}_n(k), B = B^T, C = C^T \right\}$$

Definition 2.1.7 (Special Orthogonal Lie Algebra)

Let k be a field. Let $n \in \mathbb{N}$. Define the even dimensional special orthogonal Lie algebra to be

$$\mathfrak{so}_{2n} = \left\{ \begin{pmatrix} A & B \\ C & -A^T \end{pmatrix} \in \mathfrak{gl}_{2n}(k) \mid A, B, C \in \mathfrak{gl}_n(k), B = -B^T, C = -C^T \right\}$$

Define the odd dimensional special orthogonal Lie algebra to be

$$\mathfrak{so}_{2n+1} = \left\{ \begin{pmatrix} 0 & E^T & -D^T \\ D^T & A & B \\ -E & C & -A^T \end{pmatrix} \in \mathfrak{gl}_{2n+1}(k) \mid A, B, C \in \mathfrak{gl}_n(k), D, E \in M_{n \times 1}(k), B = -B^T, C = -C^T \right\}$$

Proposition 2.1.8

Let k be a field. Let $n \in \mathbb{N} \setminus \{0\}$. Then the following are true.

- $\mathfrak{sl}_n(k)$ is a Lie sub-algebra of $\mathfrak{gl}_n(k)$ of dimension $n^2 - 1$.
- $\mathfrak{b}_n(k)$ is a Lie sub-algebra of $\mathfrak{gl}_n(k)$ of dimension $\frac{n^2+n}{2}$.

- $\mathfrak{u}_n(k)$ is a Lie sub-algebra of $\mathfrak{gl}_n(k)$ of dimension $\frac{n^2-n}{2}$.
- $\mathfrak{sp}_{2n}(k)$ is a Lie sub-algebra of dimension $2n^2 + n$.
- $\mathfrak{so}_{2n}(k)$ is a Lie sub-algebra of dimension $2n^2 + n$
- $\mathfrak{so}_{2n+1}(k)$ is a Lie sub-algebra of dimension $2n^2 + 3n$

Example 2.1.9

Write $e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, $f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$, $h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$. The structural constants of $\mathfrak{sl}_2(\mathbb{C})$ is given by

$$[e, f] = h \quad [e, h] = -2e \quad [f, h] = 2f$$

Proof We have

$$\begin{aligned} [e, f] &= ef - fe = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = h \\ [e, h] &= eh - he = \begin{pmatrix} 0 & -1 \\ 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = -2e \\ [f, h] &= fh - hf = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 \\ -1 & 0 \end{pmatrix} = 2f \end{aligned}$$

■

2.2 Nilpotency and Engel's Theorem

Example 2.2.1

Consider the following Lie algebras.

- $SL(2, \mathbb{C})$ is nilpotent.
- $b_n(\mathbb{C})$ is not nilpotent for all $n \geq 2$.

Example 2.2.2

Let $n \in \mathbb{N}$ such that $n \geq 2$. Let k be a field. Then $\mathfrak{u}_n(k)$ is nilpotent.

Proof Write $e_{i,j}$ the matrix with 1 at the (i, j) th position and 0 everywhere else. Notice that a basis for $\mathfrak{u}_2(\mathbb{C})$ is given by $\{e_{i,j} \mid 1 \leq i < j \leq n\}$. Moreover, one can compute that $[e_{i,j}, e_{k,l}] = \delta_{j,k}e_{i,l} - \delta_{i,l}e_{k,j}$.

I claim that

$$\mathfrak{u}_n(k)^p = \langle e_{i,j} \mid 1 \leq i, i+p < j \leq n \rangle$$

for all $0 \leq p \leq n-1$. We proceed by induction. The base case is trivial.

Assume the result is true for some $p \leq n-2$. Let $a, b \in \mathbb{N}$ be such that $a+p+1 < b$. Then notice that $e_{a,b} = [e_{a,a+1}, e_{a+p+1,b}] \in [\mathfrak{u}_n(k), \mathfrak{u}_n(k)^p] = \mathfrak{u}_n(k)^{p+1}$. However, notice that $e_{a,a+p+1}$ does not lie in $\mathfrak{u}_n(k)^{p+1}$. By anti-symmetry notice that we just need to consider the case $j = k$ in $[e_{i,j}, e_{k,l}]$. But $i+p < j+p < l$ implies that $l-i > p+1$. But $(a+p+1) - a = p+1$ means that no $i, l \in \mathbb{N}$ is such that $e_{a,a+p+1} = [e_{i,j}, e_{j,l}]$. Hence we are done.

Then it is clear that $\mathfrak{u}_n(k)^{n-1} = 0$. So $\mathfrak{u}_n(k)$ is nilpotent. ■

Lemma 2.2.3

Let R be a commutative ring. Let L be a Lie algebra over R . Then the following are true.

- Let M be a Lie subalgebra of L . If L is nilpotent, then M is nilpotent.
- If $L \neq 0$ is nilpotent, then $Z(L) \neq 0$
- If $L/Z(L)$ is nilpotent, then L is nilpotent.

Proof

- Let M be a Lie subalgebra of L . I claim that $M^k \subseteq L^k$ for all $k \in \mathbb{N}$. The base case $k = 0$ is clearly true. Suppose that $M^k \subseteq L^k$. Let $x \in [M, M^k] = M^{k+1}$. Then $x = [m, t]$ for some $m \in M$ and $t \in M^k$. Then $t \in L^k$. Also $m \in L$ implies that $x = [m, t] \in [L, L^k] = L^{k+1}$. Thus $M^{k+1} \subseteq L^{k+1}$. Now since L is nilpotent, there exists $n \in \mathbb{N}$ such that $L^n = 0$. Then $M^n \subseteq L^n = 0$ so that M is also nilpotent.
- Suppose that $n \in \mathbb{N}$ is the smallest natural number such that $L^n = 0$. Then $[L, L^{n-1}] = 0$. Let $x \in L$. Then for all $y \in L^{n-1}$, we have that $[x, y] = 0$. Thus $x \in Z(L)$.
- Since $L/Z(L)$ is nilpotent, there exists $n \in \mathbb{N}$ such that $(L/Z(L))^n = 0$. Let $\pi : L \rightarrow L/Z(L)$ be the quotient homomorphism. Since π is surjective, we use the above lemma to find that

$$\pi(L^n) = \pi(L)^n = \left(\frac{L}{Z(L)} \right)^n = \frac{L^n + Z(L)}{Z(L)} = 0$$

This means that $L^n \subseteq Z(L)$. It follows that $L^{n+1} = [L, L^n] \subseteq [L, Z(L)] = 0$ and we conclude. ■

Proposition 2.2.4

Let k be a field. Let V be a vector space over k . Let $L \subseteq \mathfrak{gl}(V)$ be a Lie sub-algebra such that for all $T \in L$, T is nilpotent. Then

$$V_0 = \{v \in V \mid T(v) = 0 \text{ for all } T \in L\} \neq \{0\}$$

Proof We induct on the dimension of V . When $\dim(L) = 1$, we have $L = \langle T \rangle$ for some non-zero $T \in \mathfrak{gl}(V)$. Suppose that $T^n = 0$ but $T^{n-1} \neq 0$. Suppose that $x \in V$ is such that $T^{n-1}(x) = y \neq 0$. Then for any $S \in L = \langle T \rangle$, write $S = \lambda T$ for some $\lambda \in k$. Then we have

$$S(y) = \lambda T(T^{n-1}(x)) = 0$$

since $T^n = 0$. Thus $0 \neq x \in V_0 \neq \{0\}$.

Suppose the result is true for all Lie sub-algebras of $\mathfrak{gl}(V)$ of dimension $< k$. Let L be a Lie sub-algebra of $\mathfrak{gl}(V)$ of dimension k . Let M be a maximal proper Lie sub-algebra of L . Such a Lie sub-algebra exists since L is finite dimensional and $\{0\}$ is a proper Lie sub-algebra.

Step 1: Apply inductive hypothesis to homomorphic image of M .

Define $f : M \rightarrow \mathfrak{gl}(L/M)$ by

$$f(m)(x + M) \mapsto [m, x] + M$$

I claim that f is a Lie algebra homomorphism. Such a map is well defined since $x \in M$ implies that $[m, x] \in M$. Moreover, since $[-, -]$ is bilinear, the map $f(m)$ lies in $\mathfrak{gl}(L/M)$ and that f is linear. Finally, we check that

$$\begin{aligned} [f(m_1), f(m_2)](x + M) &= (f(m_1)f(m_2) - f(m_2)f(m_1))(x + M) \\ &= f(m_1)([m_2, x] + M) - f(m_2)([m_1, x] + M) \\ &= [m_1, [m_2, x]] - [m_2, [m_1, x]] + M \\ &= -[[m_2, x], m_1] + [[m_1, x], m_2] + M \\ &= -[[m_2, x], m_1] - [[x, m_1], m_2] + M \\ &= [[m_1, m_2], x] + M && \text{(Jacobi Identity)} \\ &= f([m_1, m_2])(x + M) \end{aligned}$$

for all $m_1, m_2 \in M$ and $x \in L$. Thus f is a Lie algebra homomorphism. Let $m \in M \subseteq L$. Then m is nilpotent, and so m is ad-nilpotent by lemma 2.2.5. This means that $\text{ad}(m)^n = 0$ for some

$n \in \mathbb{N} \setminus \{0\}$. Then we have

$$f(m)^n(x + M) = \text{ad}(m)^n(x) + M = M$$

and that $f(m)$ is nilpotent. Now we have that $\dim(f(M)) \leq \dim(M) < \dim(L) = k$ and all elements of $f(M)$ are nilpotent. We can apply the induction hypothesis to conclude that there exists $x \in L \setminus M$ such that

$$x + M \in \{y \in L/M \mid T(y) = 0 \text{ for all } T \in f(M)\}$$

Step 2: Apply inductive hypothesis to M .

I claim that $M \oplus \langle x \rangle$ is a Lie sub-algebra of L . To see this, let $n_1, n_2 \in M \oplus \langle x \rangle$, which we can write as $n_1 = m_1 + \lambda_1 x$ and $n_2 = m_2 + \lambda_2 x$ for some $m_1, m_2 \in M$ and $\lambda_1, \lambda_2 \in k$. Then we check that

$$[n_1, n_2] = [m_1 + \lambda_1 x, m_2 + \lambda_2 x] = [m_1 + m_2] + \lambda_1 [x, m_2] + \lambda_2 [m_1, x] \in M$$

so that $M \oplus \langle x \rangle$ is indeed a Lie sub-algebra of L . Since M is maximal and $\dim(M \oplus \langle x \rangle) > \dim(M)$, we conclude that $M \oplus \langle x \rangle = L$. Moreover, M is an ideal of L since we can check that for any $z \in L$ and $m \in M$, we can write $z = m + \lambda x$ for $m \in M$ and $\lambda \in k$ and compute that

$$[z, m'] = [m, m'] + \lambda [x, m'] \in M$$

for any $m' \in M$. Now apply the inductive hypothesis on M to get that

$$W = \{v \in V \mid T(v) = 0 \text{ for all } T \in M\}$$

is a non-trivial subspace of L .

Step 3: Find the element in L satisfying the hypothesis.

By 3.2.3, W is an L -invariant subspace. This means that $x(w) \in W$ for all $w \in W$ and $x \in L$. Since $x \in L$, x is nilpotent. Suppose that $x^t = 0$ but $x^{t-1} \neq 0$. This means that there exists $w \in W$ such that $q = x^{t-1}(w) \neq 0$, and $x(q) = 0$. Let $T \in L$ be arbitrary. Since $L = M \oplus \langle x \rangle$, we can write $T = m + \lambda x$ for some $m \in M$ and $\lambda \in k$. Then we have

$$T(q) = m(q) + \lambda x(q) = 0$$

because $q \in W$ implies that $m(q) = 0$ for all $m \in M$. Thus $0 \neq q \in V_0$ so that we conclude. ■

Proposition 2.2.5 (Engel's Theorem for Lie Sub-algebras of $\mathfrak{gl}(V)$)

Let k be a field. Let V be a finite dimensional vector space over k . Let L be a Lie subalgebra of $\text{End}(V)$. If all $x \in L$ are nilpotent, then the following are true.

- There exists a basis for V such that all elements of L are strictly upper triangular.
- L is nilpotent.

Proof We induct on the dimension of V . When $\dim(V) = 1$, then $\mathfrak{gl}(V) \cong k$ as vector spaces. The only nilpotent element of k is 0. Thus $L = \{0\}$ and we are done.

Assume the results are true for all vector spaces V with $\dim(V) < k$. Let V be a k dimensional vector space. By prp4.2.9, there exists $v \in V$ such that $0 \neq v \in V_0$. Let $W = V/\langle v \rangle$. Define a map $\phi : L \rightarrow \mathfrak{gl}(W)$ by

$$\phi(x)(u + \langle v \rangle) = x(u) + \langle v \rangle$$

Notice that this is well defined since $u \in \langle v \rangle$ implies that $x(u) = 0$ by definition of v . Moreover, $\phi(x)$ is a well defined linear map since x is a linear map. Finally, ϕ is a linear map since addition

in $\mathfrak{gl}(V)$ is defined pointwise. It is also a Lie algebra homomorphism since

$$\begin{aligned}\phi([x, y])(u + \langle v \rangle) &= [x, y](u) + \langle v \rangle \\ &= x(y(u)) - y(x(u)) + \langle v \rangle \\ &= \phi(x)(\phi(y)(u)) - \phi(y)(\phi(x)(u)) + \langle v \rangle \\ &= [\phi(x), \phi(y)](u + \langle v \rangle)\end{aligned}$$

Hence $\phi(L)$ is a Lie sub-algebra of $\mathfrak{gl}(W)$.

Recall that all $x \in L$ are nilpotent, say $x^n = 0$. Then we have

$$\phi(x)^n(u + \langle v \rangle) = x^n(u) + \langle v \rangle = \langle v \rangle$$

so that all elements of $\phi(L)$ are nilpotent. Since $\dim(W) = \dim(V) - 1$, we can apply the induction hypothesis to get a basis $u_1 + \langle v \rangle, \dots, u_{k-1} + \langle v \rangle$ such that all elements of $\mathfrak{gl}(W)$ are strictly upper triangular. I claim that $v, u_1, \dots, u_{k-1}$ is a basis for L so that all elements of L are strictly upper triangular. Let $x \in L$. The matrix of x is given by the matrix of $x|_{\langle v \rangle}$ and the block sum of the matrix of $x|_W = \phi(x)$. By assumption $\phi(x)$ is upper triangular with respect to $u_1 + \langle v \rangle, \dots, u_{k-1} + \langle v \rangle$. Since $x(v) = 0$ by construction, we conclude that x is upper triangular with respect to the basis $v, u_1, \dots, u_{k-1}$ and so the first part is complete.

For the second part, we know that V has a basis for which all elements of L are strictly upper triangular. This means that L is a Lie subalgebra of $\mathfrak{u}_n(k)$ for $n = \dim(V)$. By 3.2.7 and 3.2.8 we conclude that L is nilpotent. ■

2.3 Solubility and Lie's Theorem

Theorem 2.3.1 (Engel's Theorem)

Let L be a finite dimensional Lie algebra. Then L is nilpotent if and only if all $x \in L$ are ad-nilpotent.

Proof Let L be nilpotent. Then $L^n = 0$ for some $n \in \mathbb{N}$. Since $\text{ad}(x)^n(y) \in L^n$ for all x and y , we conclude that x is ad-nilpotent.

Conversely, suppose that all $x \in L$ are ad-nilpotent. Consider the adjoint homomorphism $\text{ad} : L \rightarrow \mathfrak{gl}(L)$. Since all $x \in L$ are ad-nilpotent, this means that $\text{ad}(x) \in \mathfrak{gl}(L)$ is nilpotent for all $x \in L$. Then by Engel's theorem for Lie sub-algebras of $\mathfrak{gl}(V)$, we conclude that $\text{ad}(L)$ is nilpotent. Since $\text{ad}(L) \cong \frac{L}{Z(L)}$, by 3.2.8 we conclude that L is nilpotent. ■

Proposition 2.3.2

Let L be a Lie algebra of dimension 2. Then L is soluble.

Proof If L is abelian, then lmm 2.2.4 implies that L is soluble. So suppose that L is non-abelian. Then $L^{(1)} = [L, L] \neq 0$. Suppose for a contradiction that $[L, L] = L$. Let $v, w \in L$ be non-zero. Define $x = [v, w]$ and extend it to a basis $\{x, y\}$ of L . Suppose that $v = ax + by$ and $w = cx + dy$. Then we have that

$$x = [v, w] = [ax + by, cx + dy] = (ad - bc)[x, y]$$

Since $[L, L] = L$, there exists $f, g \in L$ such that $[f, g] = y$. Let $f = px + qy$ and $g = rx + sy$. Then we have that

$$y = [f, g] = [px + qy, rx + sy] = (ps - qr)[x, y]$$

Then we have that

$$\frac{1}{ad - bc}x = [x, y] = \frac{1}{(ps - qr)}y$$

which means that x and y are linearly dependent. This is a contradiction. Hence $[L, L] \neq L$.

Since $[L, L]$ is a Lie subalgebra of L , $[L, L]$ must be 1-dimensional. By the classification of 1-dimensional Lie algebras, $L^{(1)} = [L, L]$ is abelian. Hence $L^{(2)} = [L^{(1)}, L^{(1)}] = 0$. Thus L is soluble. ■

Example 2.3.3

The Lie algebras $\mathfrak{gl}_2(\mathbb{C})$ and $\mathfrak{sl}_2(\mathbb{C})$ are not soluble.

Proof Using a similar method as Example 3.2.7, recall that a basis for $\mathfrak{gl}_2(\mathbb{C})$ is given by $\{e_{1,1}, e_{1,2}, e_{2,1}, e_{2,2}\}$. Then we compute that $\mathfrak{gl}_2(\mathbb{C})^{(1)}$ is spanned by the elements

- $[e_{1,1}, e_{1,2}] = e_{1,2}$
- $[e_{1,1}, e_{2,1}] = -e_{2,1}$
- $[e_{1,1}, e_{2,2}] = 0$
- $[e_{1,2}, e_{2,1}] = e_{1,1} - e_{2,2}$
- $[e_{1,2}, e_{2,2}] = e_{1,2}$
- $[e_{2,1}, e_{2,2}] = -e_{2,1}$

So that $\mathfrak{gl}_2(\mathbb{C})^{(1)} = \mathfrak{sl}_2(\mathbb{C})$. We can then compute $\mathfrak{gl}_2(\mathbb{C})^{(2)} = \mathfrak{sl}_2(\mathbb{C})^{(1)}$ which is spanned by

- $[e_{1,2}, e_{2,1}] = e_{1,1} - e_{2,2}$
- $[e_{1,2}, e_{1,1} - e_{2,2}] = -2e_{1,2}$
- $[e_{2,1}, e_{1,1} - e_{2,2}] = e_{2,1} + e_{2,2}$

So that $\mathfrak{gl}_2(\mathbb{C})^{(2)} = \mathfrak{sl}_2(\mathbb{C})^{(1)} = \mathfrak{sl}_2(\mathbb{C})$. Since both of the derived series stabilizes, we conclude that they are not soluble. ■

Example 2.3.4

Let $n \geq 2$. Then $\mathfrak{b}_n(\mathbb{C})$ is soluble.

Proof Using a similar method as Example 3.2.7, we conclude that $\mathfrak{b}_n(\mathbb{C})^{(1)} = [\mathfrak{b}_n(\mathbb{C}), \mathfrak{b}_n(\mathbb{C})] = \mathfrak{u}_n(\mathbb{C})$. Assume that $\mathfrak{b}_n(\mathbb{C})^{(k)} = \mathfrak{u}_n(\mathbb{C})^{(k-1)}$. Then we have

$$\mathfrak{b}_n(\mathbb{C})^{(k+1)} = [\mathfrak{b}_n(\mathbb{C})^{(k)}, \mathfrak{b}_n(\mathbb{C})^{(k)}] = [\mathfrak{u}_n(\mathbb{C})^{(k-1)}, \mathfrak{u}_n(\mathbb{C})^{(k-1)}] = \mathfrak{u}_n(\mathbb{C})^{(k)}$$

So by induction we have $\mathfrak{b}_n(\mathbb{C})^{(n+1)} = \mathfrak{u}_n(\mathbb{C})^{(n)}$ for all $n \in \mathbb{N}$. Since $\mathfrak{u}_n(\mathbb{C})$ is nilpotent, it is also soluble. Hence $\mathfrak{b}_n(\mathbb{C})$ is also soluble. ■

Proposition 2.3.5

Let V be a vector space over $\mathbb{C}$. Let L be a Lie sub-algebra of $\mathfrak{gl}(V)$. If L is soluble, then there exists $0 \neq v \in V$ such that v is an eigenvector for T for all $T \in L$.

Proof We proceed by induction on the dimension of L . When $\dim(L) = 1$ it is clear. Suppose that the result is true for any Lie algebra of dimension $< n$. Let L be a Lie algebra of dimension n .

Step 1: Find an ideal M of dimension $\dim(L) - 1$ containing $[L, L]$.

Since L is soluble, $[L, L] \neq L$. Let M be a vector space containing $[L, L]$ of dimension $n - 1$. I claim that M is an ideal of L . Indeed, let $x \in L$ and $m \in M$. Then we have

$$[x, m] \in [L, L] \subseteq M$$

Let $z \in L/M$ such that $L = M \oplus \langle z \rangle$.

Step 2: Apply induction hypothesis to M .

Since $\dim(M) = \dim(L) - 1$, we can apply the inductive hypothesis to see that there exists $w \in V$ that is a common eigenvector for all $m \in M$. Let $\lambda : M \rightarrow \mathbb{C}$ be the corresponding weight of m . By 3.2.4, V_λ is L -invariant and also $\langle z \rangle$ -invariant. This means that there exists $0 \neq v \in V_\lambda$ that is an eigenvector for both z and hence $\langle z \rangle$, and M . ■

Theorem 2.3.6 (Lie's Theorem)

Let V be a vector space over $\mathbb{C}$. Let L be a soluble Lie sub-algebra of $\mathfrak{gl}(V)$. Then there exists a basis B of V such that for all $T \in L$, T is upper triangular.

Proposition 2.3.7

Let L be a Lie algebra over $\mathbb{C}$. Then L is soluble if and only if $[L, L]$ is nilpotent.

Proof Suppose that $[L, L]$ is nilpotent. Then $[L, L]$ is soluble by 3.3.5. Since $L/[L, L]$ is abelian, it is also soluble. Hence by 3.3.9, L is soluble.

Suppose that L is soluble. Then $\text{ad}(L) \leq \mathfrak{gl}(L)$ is soluble by 3.3.9. By Lie's theorem, there exists a basis B of V such that for all $T \in \text{ad}(L)$, T is upper triangular. Hence $\text{ad}(L)$ is isomorphic as Lie algebras to a Lie sub-algebra of $\mathfrak{b}_n(\mathbb{C})$. Since $\text{ad}(L)^1 = \mathfrak{b}_n(\mathbb{C})^1 = \mathfrak{u}_n(\mathbb{C})$ and the latter is nilpotent, we conclude that $\text{ad}(L)$ is nilpotent. Then $\text{ad}([L, L]) = [\text{ad}(L), \text{ad}(L)]$ implies that $\text{ad}(x)$ is nilpotent for all $x \in [L, L]$. By Engel's theorem, $[L, L]$ is nilpotent ■

2.4 Low Dimensional Lie Algebras

Lemma 2.4.1

Let L be a Lie algebra of dimension 1. Then L is abelian.

Proof If L is 1-dimensional over a field k , let x be a spanning element of L . Then for $s, t \in L$, there exists $a, b \in k$ such that $s = ax$ and $t = bx$. Now we have that

$$[ax, bx] = ab[x, x] = 0$$

Hence L is abelian. ■

Proposition 2.4.2

Let L be a Lie algebra over a field k such that $\dim(L) = 1$. Then L is isomorphic to k equipped with the trivial Lie bracket. In particular, L is abelian.

Proposition 2.4.3

Let L be a Lie algebra over a field k such that $\dim(L) = 2$. Then L is isomorphic to one of the following.

- The vector space k^2 together with the trivial Lie bracket.
- The vector space $k^2 = \langle x, y \rangle$ together with the Lie bracket defined by $[x, y] = x$.

In both cases, L is soluble.

Proof Let L be non-abelian. Then $[L, L]$ must have dimension at least 1. It must also have dimension at most 1, otherwise $[L, L] = L$ implies that L is not soluble. Thus $\dim([L, L]) = 1$. Suppose that x spans $[L, L]$. Let $y \in L$ be such that $\{x, y\}$ is a basis for L . Then $[x, y] = \lambda x$ for some $\lambda \neq 0$. Then let $y' = \frac{1}{\lambda}y$ so that $[x, y'] = x$. Hence L has a basis $\{x, y\}$ such that $[x, y] = x$ and we are done. ■

Proposition 2.4.4

Let L be a Lie algebra over $\mathbb{C}$ such that $\dim(L) = 3$. Then L is isomorphic to one of the following.

- The vector space $\mathbb{C}^3$ together with the trivial Lie bracket.
- $\mathfrak{u}_3(\mathbb{C})$. (Case $\dim([L, L]) = 1$ and $[L, L] \subseteq Z(L)$)
- The direct sum of the non-abelian $\mathbb{C}^2$ Lie algebra and $\mathbb{C}$. (Case $\dim([L, L]) = 1$ and $[L, L] \not\subseteq Z(L)$)

- $Z(L)$
- (Case $\dim([L, L]) = 2$)
 - $\mathfrak{sl}_2(\mathbb{C})$ (Case $\dim([L, L]) = 3$)

Lemma 2.4.5

Let L be a Lie algebra such that $\dim(L) = 3$. If $Z(L) \neq \emptyset$ then L is nilpotent.

Lemma 2.4.6

If L is semisimple, then $\dim(L) \geq 3$.

Proof Any Lie algebra of dimension 1 or 2 must be soluble, and cannot be semisimple. ■

Example 2.4.7

The following are true.

- $\mathfrak{gl}_2(\mathbb{C})$ is not semisimple.
- $\mathfrak{sl}_2(\mathbb{C})$ is semisimple.

Proof Since $Z(\mathfrak{gl}_2(\mathbb{C}))$ is abelian, it is soluble. Hence $Z(\mathfrak{gl}_2(\mathbb{C})) \subseteq \text{rad}(\mathfrak{gl}_2(\mathbb{C}))$. Thus $\mathfrak{gl}_2(\mathbb{C})$ is not semisimple.

I claim that

$$\text{rad}(\mathfrak{gl}_2(\mathbb{C})) = Z(\mathfrak{gl}_2(\mathbb{C})) = \{aI_2 \mid a \in \mathbb{C}\}$$

The second equality is clear. Notice that if the dimension of $\text{rad}(\mathfrak{gl}_2(\mathbb{C}))$ is 2 or 3, then $\frac{\mathfrak{gl}_2(\mathbb{C})}{\text{rad}(\mathfrak{gl}_2(\mathbb{C}))}$ has dimension 1 or 2. In both cases, we have seen that it must be soluble. This is a contradiction since $\text{rad}(\mathfrak{gl}_2(\mathbb{C}))$ being soluble implies that $\mathfrak{gl}_2(\mathbb{C})$ is soluble. Therefore $\dim_{\mathbb{C}}(\text{rad}(\mathfrak{gl}_2(\mathbb{C}))) = 1$. Since $Z(\mathfrak{gl}_2(\mathbb{C}))$ also has dimension 1, we conclude that $Z(\mathfrak{gl}_2(\mathbb{C})) = \text{rad}(\mathfrak{gl}_2(\mathbb{C}))$.

Consider the composite Lie algebra homomorphism given by the inclusion $\mathfrak{sl}_2(\mathbb{C}) \hookrightarrow \mathfrak{gl}_2(\mathbb{C})$ composed with the projection $\mathfrak{gl}_2(\mathbb{C}) \rightarrow \frac{\mathfrak{gl}_2(\mathbb{C})}{Z(\mathfrak{gl}_2(\mathbb{C}))}$. They both have dimension 3. Moreover, it is injective because if $\begin{pmatrix} a & b \\ c & -a \end{pmatrix} \in Z(\mathfrak{gl}_2(\mathbb{C}))$, then we conclude that $a = b = c = 0$. Therefore the composite map is a Lie algebra isomorphism. By the above theorem, we have that

$$\text{rad}(\mathfrak{sl}_2(\mathbb{C})) \cong \text{rad}\left(\frac{\mathfrak{gl}_2(\mathbb{C})}{Z(\mathfrak{gl}_2(\mathbb{C}))}\right) = \text{rad}\left(\frac{\mathfrak{gl}_2(\mathbb{C})}{\text{rad}(\mathfrak{gl}_2(\mathbb{C}))}\right) = 0$$

Thus $\mathfrak{sl}_2(\mathbb{C})$ is semisimple. ■

Example 2.4.8

$\mathfrak{sl}_2(\mathbb{C})$ is a simple Lie algebra.

Proof We know that $\dim_{\mathbb{C}}(\mathfrak{sl}_2(\mathbb{C})) = 3$. Let I be a proper non-zero ideal of $\mathfrak{sl}_2(\mathbb{C})$. Then I has dimension 1 or 2. As a Lie algebra I must be soluble. Then $I \subseteq \text{rad}(\mathfrak{sl}_2(\mathbb{C})) \neq \{0\}$ so that $\mathfrak{sl}_2(\mathbb{C})$ is not semisimple, a contradiction. ■

3 The Killing Form of a Lie Algebra

3.1 The Killing Form

Let $A = (a_{i,j})$ be an $n \times n$ matrix. Recall that the trace of A is defined as

$$\text{tr}(A) = \sum_{k=1}^n a_{k,k}$$

Now let $T : V \rightarrow V$ be a linear map. Then we can also define the trace of T abstractly so that any choice of representation of T with a matrix gives coinciding trace. This is also given in Linear Algebra. The formula is

$$\text{tr}(T) = \sum_{i=1}^n \langle T(e_i), e_i \rangle$$

Definition 3.1.1 (The Killing Form)

Let L be a Lie algebra over $\mathbb{C}$. Define the killing form of L to be the function

$$k_L : L \times L \rightarrow \mathbb{C}$$

given by $k_L(x, y) = \text{tr}(\text{ad}(x) \circ \text{ad}(y))$.

Lemma 3.1.2

Let L be a Lie algebra. Then the following are true.

- The killing form on L is a symmetric bilinear form.
- $k_L([x, y], z) = k_L(x, [y, z])$ for all $x, y, z \in L$.

Proof Let L be a Lie algebra over a field $\mathbb{F}$. Let $x, y \in L$. Then we have that

$$k_L(x, y) = \text{tr}(\text{ad}(x) \circ \text{ad}(y)) = \text{tr}(\text{ad}(y) \circ \text{ad}(x)) = k_L(y, x)$$

since the trace function preserve commutation. Thus k_L is symmetric. Because it is symmetric, it is sufficient to show that k_L is linear in the first variable for bilinearity. But the adjoint homomorphism is linear and composition preserves linearity. Hence k_L is bilinear.

Finally, we have that

$$\begin{aligned} k_L([x, y], z) &= \text{tr}(\text{ad}([x, y]) \circ \text{ad}(z)) \\ &= \text{tr}([\text{ad}(x), \text{ad}(y)] \circ \text{ad}(z)) && (\text{ad is a Lie algebra Hom}) \\ &= \text{tr}(\text{ad}(x) \circ [\text{ad}(y), \text{ad}(z)]) && (\text{Imm1.6.2}) \\ &= k_L(x, [y, z]) \end{aligned}$$

■

Lemma 3.1.3

Let L be a finite dimensional Lie algebra. Let I be an ideal of L . Then we have

$$k_I = k_L|_{I \times I}$$

Proof Let I be an ideal of L so that I is also a Lie subalgebra of L . Let B_I be a basis for I . Extend it to a basis B_L of L . Let $x \in I$. Then we have a Lie algebra homomorphism $\text{ad}(x) : L \rightarrow I$ since $[x, z] \in I$ for all $z \in L$. We can then represent $\text{ad}(x)$ in the basis B using the matrix

$$T = \begin{pmatrix} A_x & B_x \\ 0 & 0 \end{pmatrix}$$

where A_x is the matrix representing the linear map $\text{ad}(x)|_I : I \rightarrow I$. For any $x, y \in I$, we have that

$$\begin{aligned} k_L(x, y) &= \text{tr}(\text{ad}(x) \circ \text{ad}(y)) \\ &= \text{tr} \left(\begin{pmatrix} A_x & B_x \\ 0 & 0 \end{pmatrix} \begin{pmatrix} A_y & B_y \\ 0 & 0 \end{pmatrix} \right) \\ &= \text{tr} \left(\begin{pmatrix} A_x A_y & A_x B_y \\ 0 & 0 \end{pmatrix} \right) \\ &= \text{tr}(A_x A_y) \\ &= k_I(x, y) \end{aligned}$$

■

3.2 Cartan's Two Criteria

Lemma 3.2.1

Let V be a vector space over a field k . Let $T : V \rightarrow V$ be a linear map. Let $T = D + N$ be the Jordan-Chevalley decomposition of T . Then there exists $p, q \in \mathbb{C}[t]$ such that $p(T) = D$ and $q(T) = N$.

Proposition 3.2.2

Let V be a vector space over a field k . Let $L \leq \mathfrak{gl}(V)$ be a Lie sub-algebra. If $\text{tr}(xy) = 0$ for all $x, y \in L$, then L is soluble.

Proof Let $x \in [L, L]$. Let $x = d + n$ be the Jordan decomposition of $x \in \mathfrak{gl}(V)$. Choose a basis B for V such that d is diagonal. Then x is in its Jordan canonical form. Since $x \in [L, L]$, there exists $y_1, \dots, y_r, z_1, \dots, z_r \in L$ such that $x = \sum_{i=1}^r [y_i, z_i]$. Then we have

$$\text{tr}(\bar{d}x) = \text{tr} \left(\bar{d} \sum_{i=1}^r [y_i, z_i] \right) = \sum_{i=1}^r \text{tr}(\bar{d}[y_i, z_i]) = \sum_{i=1}^r \text{tr}([\bar{d}, y_i]z_i)$$

By 2.2.6, the Jordan decomposition of $\text{ad}(x)$ is $\text{ad}(x) = \text{ad}(d) + \text{ad}(n)$. By the above lemma and ???, there exists a polynomial $p \in \mathbb{C}[t]$ such that $p(\text{ad}(x)) = \text{ad}(\bar{d})$. Since $\text{ad}(x) : L \rightarrow L$, we have that $\text{ad}(\bar{d}) = p(\text{ad}(x)) : L \rightarrow L$. Hence $\text{ad}(\bar{d})(y_i) = [\bar{d}, y_i] \in L$. Then $z_i \in L$ and $\text{ad}(\bar{d}) \in L$ implies that $\text{tr}([\bar{d}, y_i]z_i) = 0$. Since $\text{tr}(\bar{d}x)$ is the sum of the square of the absolute values of the elements in $\bar{d}$ and it is equal to 0, we conclude that $\bar{d} = 0$ and hence $d = 0$. So that x is nilpotent.

Then x is ad-nilpotent so by Engel's theorem, $[L, L]$ is nilpotent. Then by 4.3.12, L is soluble. ■

Theorem 3.2.3 (Cartan's First Criterion)

Let L be a Lie algebra over $\mathbb{C}$. Then L is soluble if and only if $k(x, y) = 0$ for all $x \in L$ and $y \in [L, L]$.

Proof Let L be soluble. Then $\text{ad}(L) = [L, L]$ is a Lie subalgebra of L and is soluble by 4.3.9. By Lie's theorem, there exists a basis B of L such that every element of $\text{ad}(L) \leq \text{End}_{\mathbb{C}}(L)$ is upper triangular. Hence every element of $\text{ad}([L, L]) = [\text{ad}(L), \text{ad}(L)]$ is represented by a strictly upper triangular matrix since $[b_n(k), b_n(k)] = u_n(k)$. Hence $\text{tr}(\text{ad}(x) \circ \text{ad}(y)) = 0$ for all $x \in L$ and $y \in [L, L]$.

Now suppose that $\text{tr}(\text{ad}(x) \circ \text{ad}(y)) = 0$ for all $x \in L$ and $y \in [L, L]$. This means that $\text{ad}([L, L]) = [\text{ad}(L), \text{ad}(L)]$ is soluble by the above proposition. By 4.1.3 we know that $\text{ad}(L)/([\text{ad}(L), \text{ad}(L)])$ is abelian. By prp4.3.9, we conclude that $\text{ad}(L)$ is soluble. Since $Z(L)$ is abelian, $Z(L)$ is soluble. Hence by the same proposition we conclude that L is soluble since $\text{ad}(L) = L/Z(L)$. ■

Let L be a Lie algebra. The killing form allows L to be an inner product space. Recall from Linear Algebra

that the orthogonal complement of a subspace W of L is given by

$$W^\perp = \{x \in L \mid k(x, w) = 0 \text{ for all } w \in W\}$$

Lemma 3.2.4

Let L be a Lie algebra. Let I be an ideal of L . Then $I^\perp$ is an ideal of L .

Proof We already know that $I^\perp$ is a vector space. We want to show that $[x, i] \in I^\perp$ for all $x \in L$ and $i \in I^\perp$. Let $j \in I$ be arbitrary. Then we have

$$k([x, i], j) = k(x, [i, j]) = 0$$

since $[i, j] \in I$. Hence $[x, i] \in I^\perp$ and we are done. ■

Recall from Linear Algebra that a bilinear form is non-degenerate if

$$\{v \in V \mid \tau(v, w) = 0 \text{ for all } w \in V\} = \{0\}$$

Because the killing form is symmetric, this is the same as saying that the orthogonal complement $V^\perp = \{0\}$ is trivial.

Theorem 3.2.5 (Cartan's Second Criterion)

Let L be a Lie algebra over $\mathbb{C}$. Then L is semisimple if and only if k is non-degenerate.

Proof Assume that k is degenerate. Then $L^\perp \neq \{0\}$ and hence a non-trivial ideal of L . Let $x \in L^\perp$ and $y \in [L^\perp, L^\perp] \subseteq L^\perp$. Then $k(x, y) = 0$ by definition of $L^\perp$. Then $k_{L^\perp} = 0$. By Cartan's first criterion, $L^\perp$ is soluble. Hence $L^\perp \subseteq \text{rad}(L)$ and L is not semisimple.

Now suppose that L is not semisimple. Then L contains a non-trivial abelian ideal I . Let $x \in L$ and $i \in I$. Then we have that

$$(\text{ad}(i) \circ \text{ad}(x) \circ \text{ad}(i))(y) = [i, [x, [i, y]]] = 0$$

for any $y \in L$ because $[x, [i, y]] \in I$ and I is abelian. Applying $\text{ad}(x)$ on both sides show that

$$(\text{ad}(x) \circ \text{ad}(i))^2 = 0$$

Thus $\text{ad}(x) \circ \text{ad}(i)$ is nilpotent in $\text{End}(L)$. By Engel's theorem, we can find a basis of L such that $\text{ad}(x) \circ \text{ad}(i)$ is strictly upper triangular. This means that $k(x, i) = \text{tr}(\text{ad}(x) \circ \text{ad}(i)) = 0$. But this means that $I \subseteq L^\perp$ since $i \in I \subseteq L$ is such that $k(i, x) = 0$ for all $x \in L$. Since I is non-trivial, $L^\perp$ is non-trivial. Thus k is degenerate. ■

Lemma 3.2.6

Let L be a Lie algebra over $\mathbb{C}$. Let I be an ideal of L . If L is semisimple, then the following are true.

- $I \cap I^\perp = \{0\}$
- $L = I \oplus I^\perp$ as Lie algebras
- I and $I^\perp$ are semisimple

Proof

- We know that $I \cap I^\perp$ is also an ideal of L . For $x, y \in I \cap I^\perp$, $x \in I$ and $y \in I^\perp$ implies that $k_L(x, y) = 0$. Moreover, we know that $k_{I \cap I^\perp} = k_L|_{I \cap I^\perp \times I \cap I^\perp}$. Hence $k_{I \cap I^\perp} = 0$. By Cartan's first criterion, $I \cap I^\perp$ is soluble, and so $I \cap I^\perp \subseteq \text{rad}(L) = \{0\}$. Hence $I \cap I^\perp = \{0\}$.
- By the same reasoning, $V = I \cap I^\perp$ as vector spaces. But this is also true as Lie algebras because $[I, I^\perp] \subseteq I \cap I^\perp = \{0\}$.
- Assume that I is not semisimple. Then k_I is degenerate. So there exists $0 \neq i \in I$ such that

$k_I(i, x) = 0$ for all $x \in I$. Then $k_L(i, x) = k_I(i, x) = 0$ for all $x \in I$ implies that $i \in I^\perp$. Hence $i \in I \cap I^\perp = \{0\}$, a contradiction. The proof is identical for $I^\perp$ since $I^\perp$ is also an ideal of L .

Theorem 3.2.7

Let L be a Lie algebra over $\mathbb{C}$. Then L is semisimple if and only if there exists simple ideals $I_1, \dots, I_k$ of L such that

$$L = I_1 \oplus \dots \oplus I_k$$

Proof Suppose that L is semisimple. We induct on the dimension of L . We have seen that 1-dimensional and 2-dimensional Lie algebras cannot be semisimple. So we begin with the base case $\dim(L) = 3$. If L has a proper non-trivial ideal I , then I is either 1-dimensional or 2-dimensional. Then I is soluble by the classification theorems. This means that $I \subseteq \text{rad}(L)$ is non-zero. This is a contradiction since we assumed that L is semisimple. Hence L has no proper non-trivial ideals. Hence L is simple.

Now suppose the result holds for all Lie algebras of dimension $< n$. Let $\dim(L) = n$. Let I be a non-zero minimal ideal of L . By lmm 4.2.6, $L = I \oplus I^\perp$ where I and $I^\perp$ are both semisimple. If $L = I$ then we are done. If $L \neq I$ then I has dimension strictly less than n . By inductive hypothesis, there exists simple ideals $I_1, \dots, I_k$ of I and simple ideals $J_1, \dots, J_s$ such that

$$I = \bigoplus_{i=1}^k I_i \quad \text{and} \quad I^\perp = \bigoplus_{j=1}^s J_j$$

Now we want to show that I_i and J_j are ideals of L . Let $x \in L$. Then $x = a + b$ for $a \in I$ and $b \in I^\perp$. If $z \in I_i$, then

$$[x, z] = [a, z] + [b, z] = [a, z] \in I_i$$

since $[b, z] \in I \cap I^\perp = \{0\}$. Similarly, one can show that $[x, r] \in J_j$ whenever $r \in J_j$. Thus I_i and J_j are ideals of L . Hence we obtain a direct sum decomposition

$$L = I \oplus I^\perp = \bigoplus_{i=1}^k I_i \oplus \bigoplus_{j=1}^s J_j$$

Thus the induction is complete.

Now let $L = \bigoplus_{j=1}^k I_j$. Assume that $\text{rad}(L) \neq \{0\}$. This means that $R_j = \text{rad}(L) \cap I_j$ is a soluble ideal of L and I_j . Since I_j is simple, I_j is semisimple. Hence $R_j = \{0\}$ so that

$$[R, I_j] \subseteq R \cap I_j = R_j = \{0\}$$

Let $x \in R$ and $y \in L$. Then we can write y as $y = y_1 + \dots + y_k$ for $y_j \in I_j$. Since $[R, I_j] = \{0\}$, we have that $[x, y_j] = 0$ so that $[x, y] = 0$. Thus $x \in Z(L)$. This means that $R \subseteq Z(L)$. But $Z(L)$ is abelian so $Z(L) \subseteq R$. Hence $R = Z(L)$. By prp 1.5.3, we have that

$$Z(L) = \bigoplus_{j=1}^k Z(I_j) = \{0\} \oplus \dots \oplus \{0\} = \{0\}$$

Hence $R = \{0\}$ and L is semisimple.

Given a Lie algebra L over $\mathbb{C}$, the following are now equivalent.

- L is semisimple ($\text{rad}(L) = \{0\}$)
- L contains no non-zero abelian ideals.

- The killing form $k : L \times L \rightarrow L$ is non-degenerate.
- L decomposes into a finite direct sum of simple Lie algebras.

Example 3.2.8

For all $n \geq 2$, $\mathfrak{sl}_n(\mathbb{C})$ is semi-simple.

Proof Recall that a basis for $\mathfrak{sl}_n(\mathbb{C})$ is given by

$$\{e_{i,j} \mid 1 \leq i \neq j \leq n\} \cup \{e_{i,i} - e_{i+1,i+1} \mid 1 \leq i \leq n-1\}$$

Notice that the killing form can be written as

$$k_{\mathfrak{sl}_n(\mathbb{C})}(x, y) = 2n \operatorname{tr}(xy)$$

Suppose that k is degenerate. Then there exists $0 \neq x \in \mathfrak{sl}_n(\mathbb{C})$ such that $k(x, e_{i,j}) = 0$ for $i \neq j$. Write x in terms of the basis vectors $\{e_{i,j} \mid 1 \leq i, j \leq n\}$ by

$$x = \sum_{1 \leq i, j \leq n} a_{i,j} e_{i,j}$$

Then we have

$$0 = k(x, e_{k,l}) = 2n \operatorname{tr} \left(\sum_{1 \leq i, j \leq n} a_{i,j} e_{i,j} e_{k,l} \right) = 2n \operatorname{tr} \left(\sum_{1 \leq i \leq n} a_{i,k} e_{i,l} \right) = 2n a_{l,k}$$

This implies $a_{l,k} = 0$ for all $l \neq k$ and so x is diagonal. Now write $x = \sum_{i=1}^n a_i e_{i,i}$. Then we have

$$\begin{aligned} k(x, e_{k,k} - e_{k+1,k+1}) &= 2n \operatorname{tr} \left(\sum_{i=1}^n a_i e_{i,i} e_{k,k} - \sum_{i=1}^n a_i e_{i,i} e_{k+1,k+1} \right) \\ &= 2n \operatorname{tr}(a_k e_{k,k} - a_{k+1} e_{k+1,k+1}) \\ &= 2n(a_k - a_{k+1}) \end{aligned}$$

Since $k(x, e_{k,k} - e_{k+1,k+1}) = 0$, we have that $a_k = a_{k+1}$ for $1 \leq k \leq n-1$. Since the trace of x is 0, We can only have $a_1 = a_2 = \dots = a_n = 0$. Hence $x = 0$, a contradiction. By Cartan's second criterion, $\mathfrak{sl}_n(\mathbb{C})$ is semi-simple. ■

4 Special Subalgebras of a Lie Algebra

4.1 Derivations of a Semisimple Lie Algebras over $\mathbb{C}$

Proposition 4.1.1

Let L be a semisimple Lie algebra over $\mathbb{C}$. Then

$$\text{Der}_{\mathbb{C}}(L) = \{\text{ad}(x) : L \rightarrow L \mid x \in L\} = \text{ad}(L)$$

In other words, the only derivations of L is given exactly by the adjoints.

Proof We know that $\text{ad}(L)$ is an ideal of $\text{Der}_{\mathbb{F}}(L)$. By lemma 4.1.3 we know that the killing form of $\text{ad}(L)^{\perp}$ is just the restriction of $k_{\text{Der}_{\mathbb{F}}(L)}$ to the ideal. So we compute that

$$\begin{aligned} \text{ad}(L) \cap \text{ad}(L)^{\perp} &= \text{ad}(L) \cap \{\phi \in \text{Der}_{\mathbb{F}}(L) \mid k_{\text{Der}_{\mathbb{F}}(L)}(\phi, \Phi) = 0 \text{ for all } \Phi \in M\} \\ &= \{\phi \in \text{ad}(L) \mid k|_{\text{ad}(L)}(\phi, \Phi) = 0 \text{ for all } \Phi \in \text{ad}(L)\} \end{aligned}$$

But this is precisely the orthogonal complement of $\text{ad}(L)$ in $\text{ad}(L)$. Since $\text{ad}(L) \cong L$ is semisimple, by Cartan's second criterion $k|_{\text{ad}(L)}$ is non-degenerate. Hence the orthogonal complement of $\text{ad}(L)$ in $\text{ad}(L)$ is 0, and hence $\text{ad}(L) \cap \text{ad}(L)^{\perp} = \{0\}$ in $\text{Der}_{\mathbb{F}}(V)$. Let $\phi \in \text{ad}(L)^{\perp}$. For any $x \in L$, we have that $\text{ad}(\phi(x)) = [\phi, \text{ad}(x)]$ by the above lemma. Since $\text{ad}(L)^{\perp}$ and $\text{ad}(L)$ are both ideals, this element lies in $\text{ad}(L) \cap \text{ad}(L)^{\perp} = \{0\}$. Hence $\phi(x) \in \ker(\text{ad}) = \{0\}$ for all $x \in L$. This implies that ϕ is the zero map, and so $\text{ad}(L)^{\perp} = \{0\}$.

There are now two cases to consider. If $\text{Der}_{\mathbb{F}}(V)$ is degenerate, then $\text{Der}_{\mathbb{F}}(V)^{\perp} \neq \{0\}$. Then $\{0\} = \text{ad}(L)^{\perp} \supseteq \text{Der}_{\mathbb{F}}(V)^{\perp} \neq \{0\}$ is a contradiction. Then $\text{Der}_{\mathbb{F}}(V)$ must be non-degenerate. From Linear Algebra we have that $\text{Der}(L) = \text{ad}(L) \oplus \text{ad}(L)^{\perp}$ since $\text{ad}(L) \cap \text{ad}(L)^{\perp} = \{0\}$. Since $\text{ad}(L)^{\perp} = \{0\}$, we conclude that $\text{ad}(L) \oplus \{0\} = \text{Der}_{\mathbb{F}}(V)$. ■

Recall the we have seen from Linear Algebra the following fact: If $T \in GL(V)$ is a linear map on V with minimal polynomial factorizing as

$$\mu_T = (x - \lambda_1)^{a_1} \cdots (x - \lambda_k)^{a_k}$$

where the eigenvalues λ_i are distinct and $a_i \geq 1$, then V decomposes as a direct sum of T -invariant subspaces

$$V = V_1 \oplus \cdots \oplus V_k$$

where $V_i = \ker(T - \lambda_i I)^{a_i}$ is the generalized eigenspace.

Lemma 4.1.2

Let L be a Lie algebra over $\mathbb{C}$. Let $\phi \in \text{Der}_{\mathbb{C}}(L)$ be a derivation with Jordan-Chevalley decomposition given by $\phi = d_{\phi} + n_{\phi}$ as an element of $\mathfrak{gl}(L)$. Then $d_{\phi}, n_{\phi} \in \text{Der}_{\mathbb{C}}(L)$.

Proof Let λ be an eigenvalue of ϕ . Recall that the generalized eigenspace of d_{ϕ} is given by

$$L_{\lambda} = \{x \in L \mid x \in \ker(\phi - \lambda I)^n \text{ for some } n \in \mathbb{N}\}$$

Since d_{ϕ} is the diagonal part of the decomposition of ϕ , ϕ and d_{ϕ} share the same eigenvalues. The primary decomposition from Linear Algebra also yields

$$L = \bigoplus_{\lambda \text{ an eigenvalue of } \phi} L_{\lambda}$$

I claim that $[L_{\lambda}, L_{\mu}] \subseteq L_{\lambda+\mu}$. Let $x \in L_{\lambda}$ and $y \in L_{\mu}$. We induct on $k \in \mathbb{N}$ to show that

$$(\phi - (\lambda + \mu)I)^k([x, y]) = \sum_{i=0}^k \binom{k}{i} [(\phi - \lambda I)^k(x), (\phi - \mu I)^{k-i}(y)]$$

When $k = 0$, it is trivial. Suppose that it is true for some $k \in \mathbb{N}$. Then we have

$$\begin{aligned}
& (\phi - (\lambda + \mu)I)^{k+1}([x, y]) \\
&= (\phi - (\lambda + \mu)I)^k((\phi - (\lambda + \mu)I)([x, y])) \\
&= (\phi - (\lambda + \mu)I)^k([\phi - \lambda I)(x), y] + [x, (\phi - \mu I)(y)]) \\
&= \sum_{i=0}^k \binom{k}{i} [(\phi - \lambda I)^{i+1}(x), (\phi - \mu I)^{k-i}(y)] + \sum_{i=0}^k \binom{k}{i} [(\phi - \lambda I)^i(x), (\phi - \mu I)^{k+1-i}(y)] \\
&= \sum_{i=1}^{k+1} \binom{k}{i-1} [(\phi - \lambda I)^i(x), (\phi - \mu I)^{k+1-i}(y)] + \sum_{i=0}^k \binom{k}{i} [(\phi - \lambda I)^i(x), (\phi - \mu I)^{k+1-i}(y)] \\
&= [x, (\phi - \mu I)^{k+1}(y)] + \sum_{i=1}^k \left(\binom{k}{i-1} + \binom{k}{i} \right) [(\phi - \lambda I)^i(x), (\phi - \mu I)^{k+1-i}(y)] + [(\phi - \lambda I)^{k+1}(x), y] \\
&= [x, (\phi - \mu I)^{k+1}(y)] + \sum_{i=1}^k \binom{k+1}{i} [(\phi - \lambda I)^i(x), (\phi - \mu I)^{k+1-i}(y)] + [(\phi - \lambda I)^{k+1}(x), y] \\
&= \sum_{i=0}^{k+1} \binom{k+1}{i} [(\phi - \lambda I)^i(x), (\phi - \mu I)^{k+1-i}(y)]
\end{aligned}$$

so the induction is complete. Now since $x \in L_\lambda$, there exists $p \in \mathbb{N}$ such that $(\phi - \lambda I)^p(x) = 0$. Similarly, there exists $q \in \mathbb{N}$ such that $(\phi - \mu I)^q(y) = 0$. Then choose $k = p + 1$, we see that $[x, y] \in L_{\lambda+\mu}$ by the above formula.

Since ϕ and d_ϕ share the same eigenvalues and d_ϕ is diagonalizable, the above generalized eigenspaces collapses to the usual eigenspace. So every vector $v \in L_\lambda$ is an eigenvector of d_ϕ with eigenvalue λ . To check that d_ϕ is a derivation, we see that for $x \in L_\lambda$ and $y \in L_\mu$, we have

$$d_\phi([x, y]) = (\lambda + \mu)([x, y]) = [x, \mu y] + [\lambda x, y] = [x, d_\phi(y)] + [d_\phi(x), y]$$

By bilinearity this extends to all elements of L . Hence $d_\phi \in \text{Der}_{\mathbb{F}}(L)$. Finally, ϕ and d_ϕ in $\text{Der}_{\mathbb{F}}(L)$ implies that $n_\phi = \phi - d_\phi \in \text{Der}_{\mathbb{F}}(L)$ hence we are done. ■

4.2 Cartan Subalgebras

Let V be a vector space. Then any element of $\mathfrak{gl}(V)$ admits a Jordan-Chevalley decomposition using the Jordan decomposition form. This means that for any $T \in \mathfrak{gl}(V)$, there exists $D, N \in \mathfrak{gl}(V)$ such that the following are true.

- $T = D + N$.
- D is diagonalizable and N is nilpotent.
- $DN = ND$.

However let $L \leq \mathfrak{gl}(V)$ be a Lie subalgebra. Then consider an element $T \in L$. There is no guarantee that components of the Jordan-Chevalley decomposition lies in L . Therefore we have the notion of abstract Jordan decomposition.

Definition 4.2.1 (Abstract Jordan Decomposition)

Let L be a semisimple Lie algebra over $\mathbb{C}$. Let $x \in L$. Define an abstract Jordan decomposition of x to be a pair of elements $d, n \in L$ such that the following are true.

- $x = d + n$
- $\text{ad}(d) \in \text{End}_{\mathbb{C}}(L)$ is diagonalizable and $\text{ad}(n) \in \text{End}_{\mathbb{C}}(L)$ is nilpotent.
- $[d, n] = 0$

Lemma 4.2.2

Let L be a semisimple Lie algebra over $\mathbb{C}$. Let $x \in L$ be an element. Then there exists a unique abstract Jordan decomposition of x .

Proof Let $x \in L$. Since $\text{ad}(x) \in \mathfrak{gl}(L)$, there exists a unique Jordan-Chevalley decomposition

$$\text{ad}(x) = D + N$$

for some diagonalizable map D and nilpotent map N , such that they commute. By 5.1.5, we know that $\text{ad}(x) \in \text{ad}(L) = \text{Der}_{\mathbb{C}}(L)$ implies that $D, N \in \text{Der}_{\mathbb{C}}(L)$. This means that there exists $d, n \in L$ such that $D = \text{ad}(d)$ and $N = \text{ad}(n)$. Since ad is a linear map, we have that $\text{ad}(x) = \text{ad}(d) + \text{ad}(n)$ implies that $\text{ad}(x) = \text{ad}(d + n)$. Since L and $\text{ad}(L)$ are isomorphic (L is semisimple over $\mathbb{C}$). we thus have $x = d + n$. It remains to show that $[d, n] = 0$. We have that

$$\text{ad}([d, n]) = [\text{ad}(d), \text{ad}(n)] = 0$$

Since ad is an isomorphism, we have $[d, n] = 0$. ■

Proposition 4.2.3

Let L be a semisimple Lie algebra over $\mathbb{C}$. Let V be a vector space over $\mathbb{C}$. Let $\phi : L \rightarrow \mathfrak{gl}(V)$ be a Lie algebra homomorphism. Let $x = d + n$ be the abstract Jordan decomposition of $x \in L$. Then

$$\phi(x) = \phi(d) + \phi(n)$$

is the Jordan-Chevalley decomposition of $\phi(x)$.

Definition 4.2.4 (Cartan Subalgebra)

Let L be a Lie algebra. Let $H \leq L$ be a Lie subalgebra of L . We say that H is a Cartan subalgebra of L if the following are true.

- H is abelian.
- For each $h \in H$, $\text{ad}(h)$ is diagonalizable.
- H is maximal with respect to these two properties.

Lemma 4.2.5

Let L be a semisimple Lie algebra over $\mathbb{C}$. Then there exists a non-trivial Cartan subalgebra.

Proof If for all $x \in L$, the diagonalizable part of the abstract Jordan decomposition is 0, then x is ad -nilpotent. By Engel's theorem, L is nilpotent. Then L is soluble and so $\text{rad}(L) \neq \{0\}$, a contradiction.

Hence there exists at least one element $x \in L$ whose diagonalizable part is non-zero. Let the diagonalizable part be h . Then $\mathbb{C}\langle h \rangle$ is an abelian subalgebra of L consisting entirely of diagonalizable elements. Hence the set of all abelian subalgebras consisting only of diagonalizable elements is non-empty, and has a maximal element. ■

5 Representation Theory of Lie Algebras

5.1 Weyl's Theorem

Theorem 5.1.1 (Weyl's Theorem)

Let L be a semi-simple Lie algebra over $\mathbb{C}$. Let V be an L -module. Then V is completely reducible.

Corollary 5.1.2

Let L be a semi-simple Lie algebra over $\mathbb{C}$. Let V be an L -module such that V is a finite dimensional vector space. Then there exists irreducible V -submodules $W_1, \dots, W_n$ such that

$$V = \bigoplus_{i=1}^n W_i$$

5.2 Representations of Lie Algebras

Definition 5.2.1 (Representations of a Lie Algebra)

Let L be a Lie algebra. Let V be a vector space. A representation of L is a Lie algebra homomorphism

$$\rho : L \rightarrow \mathfrak{gl}(V)$$

Proposition 5.2.2

Let L be a Lie algebra over k . Then representations of L and L -modules are in bijection

$$\{V \text{ an } L\text{-module}\} \xrightarrow{1:1} \{L \rightarrow \mathfrak{gl}(V) \text{ a Lie algebra representation}\}$$

This bijection is given by sending an L -module $\cdot : L \times V \rightarrow V$ to the Lie algebra homomorphism $\phi : L \rightarrow \mathfrak{gl}(V)$ defined by $\phi(l)(v) = l \cdot v$.

These two ways to think about the same thing is natural. Recall that a representation of a group can be thought of as either group homomorphism $G \rightarrow GL(V)$ or a $k[G]$ -module.

Idea???? This bijection is a ???-homomorphism.

Lemma 5.2.3

Let L be a Lie algebra over k . Then above bijection restricts to a bijection:

$$\{W \text{ an } L\text{-submodule of } L\} \xrightarrow{1:1} \{\text{Ideals of } L\}$$

Proof Let W be an L -submodule of L . Then W is a vector subspace of L . Moreover, for any $x \in L$ and $w \in W$, we have $[x, w] = x \cdot w \in W$ so that W is an ideal of L .

Conversely, let I be an ideal of L . Then I is a vector subspace of L . Moreover, for any $x \in L$ and $i \in I$, we have that $x \cdot i = [x, i] \in I$. Hence I is an L -submodule of L . ■

5.3 The Case of $\mathfrak{sl}(2, \mathbb{C})$

We wish to classify all $\mathfrak{sl}_2(\mathbb{C})$ -modules. Since $\mathfrak{sl}_2(\mathbb{C})$ is semisimple, Weyl's theorem implies that it suffices to classify all irreducible $\mathfrak{sl}_2(\mathbb{C})$ -modules.

Recall that the polynomial ring $\mathbb{C}[x, y]$ is graded by total degree. We write $\mathbb{C}[x, y]_d$ for the d th homogeneous component of the graded ring.

A basis for $\mathfrak{sl}_2(\mathbb{C})$ is given by

$$\mathfrak{sl}_2(\mathbb{C}) = \mathbb{C} \left\langle e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right\rangle$$

Proposition 5.3.1

Let $d \in \mathbb{N}$. The map $\phi_d : \mathfrak{sl}_2(\mathbb{C}) \rightarrow \mathfrak{gl}(\mathbb{C}[x, y]_d)$ defined on basis vectors by

$$\phi_d(e) = x \frac{\partial}{\partial y} \quad \text{and} \quad \phi_d(f) = y \frac{\partial}{\partial x} \quad \text{and} \quad \phi_d(h) = x \frac{\partial}{\partial x} - y \frac{\partial}{\partial y}$$

is an irreducible representation of $\mathfrak{sl}_2(\mathbb{C})$ (Equivalently $\mathbb{C}[x, y]_d$ is an irreducible $\mathfrak{sl}_2(\mathbb{C})$ -module).

Proof Linearity of ϕ_d is clear since $\phi_d(e), \phi_d(f)$ and $\phi_d(h)$ are linear combinations of partial derivatives. Also, we have

$$\begin{aligned} [\phi_d(e), \phi_d(f)] &= \phi_d(e) \circ \phi_d(f) - \phi_d(f) \circ \phi_d(e) \\ &= \phi_d(e) \left(y \frac{\partial}{\partial x} \right) - \phi_d(f) \left(x \frac{\partial}{\partial y} \right) \\ &= x \frac{\partial}{\partial x} + xy \frac{\partial^2}{\partial y \partial x} - y \frac{\partial}{\partial y} - xy \frac{\partial^2}{\partial x \partial y} \\ &= \phi_d(h) \\ &= \phi_d([e, f]) \end{aligned}$$

Similarly, we have

$$\begin{aligned} [\phi_d(e), \phi_d(h)] &= \phi_d(e) \circ \phi_d(h) - \phi_d(h) \circ \phi_d(e) \\ &= \phi_d(e) \left(x \frac{\partial}{\partial x} - y \frac{\partial}{\partial y} \right) - \phi_d(h) \left(x \frac{\partial}{\partial y} \right) \\ &= x^2 \frac{\partial^2}{\partial y \partial x} - x \frac{\partial}{\partial y} - xy \frac{\partial^2}{\partial y^2} - x \frac{\partial}{\partial y} - x^2 \frac{\partial^2}{\partial x \partial y} + xy \frac{\partial^2}{\partial y^2} \\ &= -x \frac{\partial}{\partial y} - x \frac{\partial}{\partial y} \\ &= \phi_d(-2e) \\ &= \phi_d([e, h]) \end{aligned}$$

The proof that $[\phi_d(f), \phi_d(h)] = \phi_d([f, h])$ is similar. Hence ϕ_d is a representation of $\mathfrak{sl}_2(\mathbb{C})$.

We are left to show that ϕ_d is an irreducible representation. Using the basis $\{x^i y^{d-i} \mid 0 \leq i \leq d\}$, we see that the matrix representation of $\phi_d(e), \phi_d(f), \phi_d(h)$ is given by

$$\phi_d(e) = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & d \\ 0 & 0 & 0 & \cdots & 0 \end{pmatrix} \quad \phi_d(f) = \begin{pmatrix} 0 & 0 & \cdots & 0 & 0 \\ d & 0 & \cdots & 0 & 0 \\ 0 & d-1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{pmatrix}$$

and

$$\phi_d(h) = \begin{pmatrix} d & 0 & \cdots & 0 & 0 \\ 0 & d-2 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & -(d-2) & 0 \\ 0 & 0 & \cdots & 0 & -d \end{pmatrix}$$

Let $U \subseteq \mathfrak{gl}(\mathbb{C}[x, y]_d)$ be a sub-module of $\mathfrak{sl}_2(\mathbb{C})$. Then $\phi_d(h)(U) \subseteq U$. Since $\phi_d(h)$ is diagonal, $\phi_d(h)|_U$ is also diagonalizable. Since all eigenvalues of $\phi_d(h)$ are different, all its eigenspaces are

one dimensional and is spanned by $x^i y^{d-i}$ for each i . Hence U contains some monomial $x^i y^{d-i}$. I claim that U contains all other monomials and hence $U = \mathfrak{gl}_2(\mathbb{C}[x, y]_d)$. Indeed, if $x^i y^{d-i} \in U$, then $\phi_d\left(\frac{f}{d-i}\right)(x^i y^{d-i}) = x^{i-1} y^{d-i+1} \in U$. Hence $x^i y^{d-i}, \dots, x^{i-1} y^{d-i+1}, \dots, y^d \in U$. Then $\phi_d(e^{d+1}) \in \mathbb{C}\langle x^d \rangle$. Hence $x^d \in U$ and so all monomials lie in U . ■

Lemma 5.3.2

Let $p : \mathfrak{sl}_2(\mathbb{C}) \rightarrow \mathfrak{gl}(V)$ be a representation. Let $v \in V$ be an eigenvector of $p(h)$ with eigenvalue λ . Then the following are true.

- Either $p(e)(v) = 0$ or $p(e)(v)$ is an eigenvector of $p(h)$ with eigenvalue $\lambda + 2$.
- Either $p(f)(v) = 0$ or $p(f)(v)$ is an eigenvector of $p(h)$ with eigenvalue $\lambda - 2$.

Proof

- If $p(e)(v) \neq 0$, then we have

$$p(h)(p(e)(v)) = p([h, e])(v) + p(e)(p(h)(v)) = p(2e)(v) + p(e)(\lambda v) = (\lambda + 2)p(e)(v)$$

- If $p(f)(v) \neq 0$, then we have

$$p(h)(p(f)(v)) = p([h, f])(v) + p(f)(p(h)(v)) = p(-2f)(v) + p(f)(\lambda v) = (\lambda - 2)p(f)(v)$$

Lemma 5.3.3

Let $p : \mathfrak{sl}_2(\mathbb{C}) \rightarrow \mathfrak{gl}(V)$ be a finite dimensional representation. Then there exists an eigenvector $v \in V$ for $p(h)$ such that the following are true.

- $p(e)(v) = 0$
- $p(f)^d(v) \neq 0$ and $p(f)^{d+1}(v) = 0$ for some $d \in \mathbb{N}$.

Proof Since V is a vector space over $\mathbb{C}$, $p(h)$ must contain at least one eigenvector w with eigenvalue λ . Consider the set $\{p(e)^n(w) \mid n \in \mathbb{N}\}$. This set must be finite, otherwise by the above lemma $p(h)$ has an infinite number of eigenvalues, and hence infinitely many eigenvectors which are linearly independent. Hence there exists the smallest $k \in \mathbb{N}$ such that $p(e)^k(w) \neq 0$ but $p(e)^{k+1}(w) = 0$. Setting $v = p(e)^{k+1}(w)$ gives us the first part. The second part is similar. ■

Theorem 5.3.4 (Classification of Irreducible Representations of $\mathfrak{sl}_2(\mathbb{C})$)

Let V be an irreducible $\mathfrak{sl}_2(\mathbb{C})$ -module. Then V is isomorphic to $\mathbb{C}[x, y]_d$ as $\mathfrak{sl}_2(\mathbb{C})$ -modules for some $d \in \mathbb{N}$.

Proof Let $v \in V$ be an eigenvector with the properties in the above lemma. I claim that $\{v, \dots, f^d \cdot v\}$ is a basis for V , and that v has eigenvalue $\lambda = d$. We know that they are linearly independent since they have different eigenvalues. By construction, U is invariant under f and h . It suffices to show that the basis vectors are invariant under e . We proceed by induction. When $i = 0$, $e \cdot (f^i \cdot v) = e \cdot v = 0$ by the above lemma. Suppose that $e \cdot (f^i \cdot v) \in U$ for some i . Then we have

$$e \cdot (f^{i+1} \cdot v) = [e, f] \cdot (f^i \cdot v) + f \cdot (e \cdot (f^i \cdot v)) = h \cdot (f^i \cdot v) + f \cdot (e \cdot (f^i \cdot v))$$

The first term lies in U since U is invariant under h and f . The second term also lies in U since U is invariant under f by induction hypothesis, U is invariant under e . Hence U is a sub-module of V . But V is an irreducible module. Hence $U = V$.

Now $\{v, \dots, f^d \cdot v\}$ are distinct eigenvectors of h because they have distinct eigenvalues. The matrix of $h \cdot -$ with respect to the basis is diagonal, and its trace is

$$\text{tr}(h \cdot -) = \lambda + (\lambda - 2) + \dots + (2 - \lambda) + (-\lambda) = (d + 1)(\lambda - d)$$

by lemma 7.4.2. Since $h \cdot - = [e \cdot -, f \cdot -] \in \mathfrak{gl}(V)$ is a commutator of some matrices, $h \cdot -$ must have zero trace. Hence $\lambda = d$.

Define $\Phi : V \rightarrow \mathbb{C}[x, y]_d$ by the formula

$$\Phi(f^i \cdot v) = \phi_d(f)^i(x^d) = f^i \cdot x^d = (d - i - 1)! x^{d-i} y^i$$

Clearly it sends the basis to basis, hence it is a vector space isomorphism. It remains to show that it is a $\mathfrak{sl}_2(\mathbb{C})$ -module homomorphism. Clearly we have

$$\Phi(f \cdot (f^i \cdot v)) = f^{i+1} \cdot (x^d) = f \cdot \Phi(f^i \cdot v)$$

Also, we have

$$\Phi(h \cdot (f^i \cdot v)) = \Phi((\lambda - 2i)(f^i \cdot v)) = (\lambda - 2i)f^i(v) = h \cdot (f^i \cdot v) = h \cdot \Phi(f^i \cdot v)$$

and

$$\begin{aligned} \Phi(e \cdot (f^i \cdot v)) &= \Phi([e, f] \cdot (f^{i-1} \cdot v) + f \cdot (e \cdot (f^{i-1} \cdot v))) \\ &= \Phi(h \cdot (f^{i-1} \cdot v) + f \cdot (e \cdot (f^{i-1} \cdot v))) \\ &= h \cdot \Phi(f^{i-1} \cdot v) + f \cdot \Phi(e \cdot (f^{i-1} \cdot v)) \\ &= h \cdot \Phi(f^{i-1} \cdot v) + f \cdot (e \cdot \Phi(f^{i-1} \cdot v)) && \text{(Induction)} \\ &= [e, f] \cdot \Phi(f^{i-1} \cdot v) + f \cdot (e \cdot \Phi(f^{i-1} \cdot v)) \\ &= e \cdot \Phi(f^i \cdot v) \end{aligned}$$

by induction. Hence we are done. ■

6 Classification of Semisimple Lie Algebras over $\mathbb{C}$

6.1 Root Space Decomposition

Let V be a vector space. We denote the dual space of V by V^* .

Let L be semisimple over $\mathbb{C}$. Let H be a Cartan subalgebra. By lmm 5.1.2, we can choose a basis $v_1, \dots, v_m$ of $GL(V)$ such that $\text{ad}(h)$ is diagonal for all $h \in H$. In particular, such a basis consists of common eigenvectors of $\text{ad}(h)$. Fix such a common eigenvector v and write its eigenvalue by $\alpha(\text{ad}(h))$ (dependent on h). Then we have

$$\text{ad}(h)(v) = \alpha(\text{ad}(h))(v)$$

and in particular $\alpha : \text{ad}(H) \rightarrow \mathbb{C}$ is a weight of $\text{ad}(H)$. Since L is semisimple, we have an isomorphism $\text{ad}(H) \cong H$. So we can think of the weight as an element

$$\alpha : H \cong \text{ad}(H) \rightarrow \mathbb{C}$$

of the dual space H^* . This motivates the following definition.

Definition 6.1.1 (Roots of Lie Algebra relative to a Cartan Subalgebra)

Let L be a semi-simple Lie algebra over $\mathbb{C}$. Let $H \leq L$ be a Cartan subalgebra of L . A root of L relative to H is an element $\alpha \in H^*$ such that $\alpha \neq 0$ and the weight space

$$L_\alpha = \{x \in L \mid [h, x] = \alpha(h)(x) \text{ for all } h \in H\}$$

is non-zero. In this case, we call L_α the root space of α .

For $x \in L_\alpha$, notice that x is a common eigenvector for $\text{ad}(h)$ for all $h \in H$.

Definition 6.1.2 (The Set of Roots relative to a Cartan Subalgebra)

Let L be a semi-simple Lie algebra over $\mathbb{C}$. Let $H \leq L$ be a Cartan subalgebra of L . Define the set of roots of L relative to H to be

$$\Phi = \{\alpha \in H^* \mid \alpha \text{ is a root of } L \text{ with respect to } H\}$$

Note that Φ is finite if L is finite dimensional.

Lemma 6.1.3

Let V be a vector space over k . Suppose that $T_1, \dots, T_n \in GL(V)$ is diagonalizable. Then there exists a basis of $GL(V)$ such that $T_1, \dots, T_n$ are diagonal if and only if $T_1, \dots, T_n$ pairwise commute. In particular, the basis are precisely common eigenvectors of $T_1, \dots, T_n$.

Lemma 6.1.4

Let L be a semisimple Lie algebra over $\mathbb{C}$. Let H be a Cartan sub-algebra of L . Let Φ be the set of roots of L relative to H . Then there is a decomposition of direct sums

$$L = L_0 \oplus \bigoplus_{\alpha \in \Phi} L_\alpha$$

given by the primary decomposition (eigenspace decomposition) of any $\text{ad}(h) \in \text{ad}(H)$.

Proof Let $h \in H$. Recall that $\text{ad}(h)$ is diagonalizable by definition. By the primary decomposition theorem (or eigenspace decomposition), we can decompose L as one-dimensional invariant subspaces:

$$L = \bigoplus_{i=1}^r \ker(\text{ad}(h) - \lambda_i(h)I)$$

where $\lambda_1(h), \dots, \lambda_r(h)$ is the complete list of distinct eigenvalues of $\text{ad}(h)$.

Since H is abelian, elements of H pairwise commute. By the above lemma, all elements of $\text{ad}(H)$ share a common set of eigenvectors. In particular, for any $h' \in H$, $\text{ad}(h)$ we have $\ker(\text{ad}(h) - \lambda_i(h)I) = \ker(\text{ad}(h') - \lambda_i(h')I)$. Then the linear functions λ_i are precisely roots of L relative to H , and so we can rewrite the decomposition as

$$L = L_0 \oplus \bigoplus_{\alpha \in \Phi} L_\alpha$$

■

Proposition 6.1.5

Let L be a semisimple Lie algebra over $\mathbb{C}$. Let $H \leq L$ be a Cartan sub-algebra of L . Let α, β be roots of L relative to H . Then the following are true.

- $[L_\alpha, L_\beta] \subseteq L_{\alpha+\beta}$
- If $\alpha + \beta \neq 0$, then $k(L_\alpha, L_\beta) = 0$ where k is the killing form of L .
- $L_0 \cap L_0^\perp = \{0\}$. In particular, $k|_{L_0}$ is non-degenerate.

Proof

- Let $x \in L_\alpha$ and $y \in L_\beta$. Then we have

$$\begin{aligned} [h, [x, y]] &= -[[x, y], h] \\ &= [[y, h], x] + [[h, x], y] \\ &= -[\beta(h)y, x] + [\alpha(h)x, y] \\ &= \alpha(h)[x, y] + \beta(h)[x, y] \\ &= (\alpha + \beta)(h)[x, y] \end{aligned}$$

so that $[x, y] \in L_{\alpha+\beta}$.

- Let $x \in L_\alpha$ and $y \in L_\beta$. Since $\alpha + \beta \neq 0$, there exists $h \in H$ such that $\alpha(h) + \beta(h) \neq 0$. Then

$$(\alpha + \beta)(h)k(x, y) = k(\alpha(h)x, \beta(h)y) = k([h, x], [h, y]) = -k([x, h], [h, y]) = -k(x, [h, [h, y]]) = 0$$

implies that $k(x, y) = 0$.

- Assume that $a \in L_0 \cap L_0^\perp$. Then $k(a, b) = 0$ for all $b \in L_0$. For any $v \in L$, we can use the above decomposition to get $v = v_0 + \sum_{\alpha \in \Phi} v_\alpha$ for $v_0 \in L_0$ and $v_\alpha \in L_\alpha$. Then we have

$$k(a, v) = k(a, v_0) + \sum_{\alpha \in \Phi} k(a, v_\alpha) = 0$$

since $a \in L_0$ and $v_\alpha \in L_\alpha$ implies $k(a, v_\alpha) = 0$ by the above. This proves that $a \in L_0^\perp$. Since L is semisimple, k is non-degenerate so that $L_0^\perp = \{0\}$. Hence $a = 0$ and $L_0 \cap L_0^\perp = \{0\}$.

■

Lemma 6.1.6

Let L be a semisimple Lie algebra over $\mathbb{C}$. Let $H \leq L$ be a Cartan sub-algebra of L . Let Φ be the set of roots of L relative to H . Then the following are true.

- For all $0 \neq h \in H$, there exists $\alpha \in \Phi$ such that $\alpha(h) \neq 0$.
- $H^* = \mathbb{C}\langle\Phi\rangle$.
- If $\alpha \in \Phi$, then $-\alpha \in \Phi$.

Proof Suppose that $-\alpha \notin \Phi$. Then $k(L_\alpha, L_\beta) = 0$ for all $\beta \in \Phi$. Hence $L_\alpha \subseteq L^\perp = \{0\}$. This is a contradiction since L_α is assumed to be non-zero. ■

Proposition 6.1.7

Let L be a semisimple Lie algebra over $\mathbb{C}$. Then we have

$$H = C_L(H) = C_L(h_0)$$

for any $h_0 \in H$ such that $\dim(C_L(h_0)) \leq \dim(C_L(H))$ for all $h \in H$.

Proof It is clear that $H \subseteq C_L(H) \subseteq C_L(h_0)$ since H is abelian.

Suppose for a contradiction that $a \in H$ but $a \notin Z(C_L(h_0))$. Since $C_L(a) \cap C_L(h_0)$ is abelian, choose a basis $\{e_1, \dots, e_k\}$ of common eigenvectors for $C_L(a) \cap C_L(h_0)$. ■

Theorem 6.1.8 (Root Space Decomposition)

Let L be a semisimple Lie algebra over $\mathbb{C}$. Let H be a Cartan sub-algebra of L . Let Φ be the set of roots of L relative to H . Then there is a decomposition of direct sums

$$L = H \oplus \bigoplus_{\alpha \in \Phi} L_\alpha$$

given by the primary decomposition (eigenspace decomposition) of any $\text{ad}(h) \in \text{ad}(H)$.

Proof Since $L_0 = C_L(H) = H$, we conclude by 9.1.5. ■

Notice that 9.1.6 then implies that the killing form on H restricted to L is non-degenerate.

Example 6.1.9

A Cartan sub-algebra of $\mathfrak{sl}_2(\mathbb{C})$ is $\mathbb{C}\langle h \rangle$, and we obtain a decomposition

$$\mathfrak{sl}_2(\mathbb{C}) = \mathbb{C}\langle h \rangle \oplus L_\alpha \oplus L_{-\alpha}$$

where $\alpha(h) = 2$.

Proof Notice that since $\mathbb{C}\langle h \rangle$ is one-dimensional, it is abelian and is contained in some Cartan subalgebra of $\mathfrak{sl}_2(\mathbb{C})$. Since $[e, h] = -2e$ and $[f, h] = 2f$ we see that the Cartan subalgebra containing h is $\mathbb{C}\langle h \rangle$. Moreover, from the two equations we see that $\alpha \in H^*$ defined by $\alpha(h) = 2$ is a root of $\mathfrak{sl}_2(\mathbb{C})$ since $L_\alpha \supseteq \mathbb{C}\langle e \rangle$ and $L_{-\alpha} \supseteq \mathbb{C}\langle f \rangle$. These make up at least a three dimensional subspace of $\mathfrak{sl}_2(\mathbb{C})$, hence we deduce the desired decomposition. ■

6.2 Copies of $\mathfrak{sl}_2(\mathbb{C})$ as Sub-algebras

Let V be a vector space over $\mathbb{C}$. Let $\tau : V \rightarrow V \rightarrow \mathbb{C}$ be a bilinear map. Recall that τ is non-degenerate if and only if $v \mapsto \tau(v, -)$ is an isomorphism between V and V^* . In this case, every linear map $T \in V^*$ determines a unique element $v \in V$ such that $T(-) = \tau(v, -)$.

Applying the above, we know that the killing form on H is non-degenerate, so any root $\alpha \in \Phi \subseteq H^*$ can be represented using the killing form.

Lemma 6.2.1

Let L be a semisimple Lie algebra over $\mathbb{C}$. Let H be a Cartan sub-algebra of L . Let Φ be a set of roots of L relative to H . Let $\alpha \in \Phi$. There exists

$$t_\alpha \in H \setminus \{0\}$$

such that the following are true.

- $\alpha(-) = k(t_\alpha, -) : H \rightarrow \mathbb{C}$ as elements of H^* .
- $[x, y] = k(x, y)t_\alpha$ for all $x \in L_\alpha$ and $y \in L_{-\alpha}$.
- $[L_\alpha, L_{-\alpha}] = \mathbb{C}\langle t_\alpha \rangle$.
- $\alpha(t_\alpha) \neq 0$.

$$\bullet t_\alpha = -t_{-\alpha}$$

Proof By 10.1.6, $k_H = k_L|_{H \times H}$ is non-degenerate. By Linear Algebra, there is an isomorphism $H \cong H^*$ given by $h \mapsto k_H(h, -)$. Since $\alpha \in H^*$, there exists $t_\alpha \in H$ such that $\alpha(-) = k_H(t_\alpha, -)$, proving the first item. t_α is non-zero otherwise $k_H(t_\alpha, -) = \alpha(-)$ is the zero map.

Since $x \in L_\alpha$ and $y \in L_{-\alpha}$, we know that $[x, y] \in L_0 = H$. Since $\alpha \neq 0$, there exists $t_\alpha \in H \setminus \{0\}$ such that $\alpha(t_\alpha) \neq 0$. Then for any $h \in H$ we have

$$k(h, [x, y]) = k([h, x], y) = k(\alpha(h)x, y) = \alpha(h)k(x, y) = k(t_\alpha, h)k(x, y)$$

Similarly, we compute that

$$k(h, k(x, y)t_\alpha) = k(x, y)k(t_\alpha, h)$$

This means that $k(h, [x, y] - k(x, y)t_\alpha) = 0$ for all $h \in H$. Since $h, [x, y] - k(x, y)t_\alpha \in H$, this implies that $[x, y] - k(x, y)t_\alpha \in H^\perp$. Proposition 10.1.6 implies that $H \cap H^\perp = \{0\}$. Hence $[x, y] = k(x, y)t_\alpha$, proving the second item.

For arbitrary $x \in L_\alpha$ and $y \in L_{-\alpha}$ we proved that $[x, y] \in \mathbb{C}\langle t_\alpha \rangle$. Hence $[L_\alpha, L_{-\alpha}] \subseteq \mathbb{C}\langle t_\alpha \rangle$. It remains to show that there exists $x \in L_\alpha$ and $y \in L_{-\alpha}$ such that $[x, y] \neq 0$. If this is not the case, then $[x, y] = 0$ for some x and y . Then $k(x, y) = 0$. By 10.1.6, $k(L_\alpha, L_\beta) = 0$ for $\alpha \neq -\beta$. This shows that $k(x, z) = 0$ for all $z \in L$. Hence $x \in L^\perp = \{0\}$. Then $L_\alpha = \{0\}$ is a contradiction. This proves the third item.

For the fourth item, suppose for a contradiction that $\alpha(t_\alpha) = 0$. For all $x \in L_\alpha$, we have $[t_\alpha, x] = \alpha(t_\alpha)x = 0$. Similarly, for all $y \in L_{-\alpha}$ we have $[t_\alpha, y] = 0$. Then $\alpha([x, y]) = k(x, y)\alpha(t_\alpha) = 0$. Hence $t_\alpha \subseteq Z(\mathbb{C}\langle x, y, [x, y] \rangle)$. By 4.4.4, $\mathbb{C}\langle x, y, [x, y] \rangle$ is nilpotent and hence soluble. Since L is semisimple, ad is an isomorphism. Hence $\text{ad}(\mathbb{C}\langle x, y, [x, y] \rangle) \cong \mathbb{C}\langle x, y, [x, y] \rangle$. By Lie's theorem, there exists a basis such that every element of $\text{ad}(\mathbb{C}\langle x, y, [x, y] \rangle)$ is upper triangular. Since $t_\alpha \in [\mathbb{C}\langle x, y, [x, y] \rangle, \mathbb{C}\langle x, y, [x, y] \rangle]$, $\text{ad}(t_\alpha)$ is strictly upper triangular. Thus $\text{ad}(t_\alpha)$ is nilpotent and so t_α is nilpotent. This is a contradiction since every element of H has non-zero diagonalizable part.

For any $h \in H$, $k(t_\alpha, h) = \alpha(h)$. So we have

$$k(t_\alpha + t_{-\alpha}, h) = k(t_\alpha, h) + k(t_{-\alpha}, h) = \alpha(h) - \alpha(h) = 0$$

Since this is true for all $h \in H$, we have that $t_\alpha + t_{-\alpha} \in H^\perp$. Since $H \cap H^\perp = \{0\}$, we conclude. ■

Definition 6.2.2 (Lie Sub-algebra associated to a Root)

Let L be a semisimple Lie algebra over $\mathbb{C}$. Let α be a root of L . Let $x \in L_\alpha$ and $y \in L_{-\alpha}$ be non-zero. Define the Lie sub-algebra associated to α by

$$M_\alpha = \mathbb{C}\langle x, y, [x, y] = k(x, y)t_\alpha \rangle$$

Proposition 6.2.3

Let L be a semisimple Lie algebra over $\mathbb{C}$. Let α be a root of L . Then the following are true. Denote

$$e_\alpha = x \quad \text{and} \quad f_\alpha = \frac{2}{k(t_\alpha, t_\alpha)k(x, y)}y \quad \text{and} \quad h_\alpha = \frac{2}{k(t_\alpha, t_\alpha)}t_\alpha$$

Then the following are true.

- $h_\alpha = -h_{-\alpha}$.
- $\alpha(h_\alpha) = 2$.
- There is a Lie algebra isomorphism

$$M_\alpha \cong \mathbb{C}\langle e_\alpha, f_\alpha, h_\alpha \rangle$$

Proposition 6.2.4

Let L be a semisimple Lie algebra over $\mathbb{C}$. Let α be a root of L . Then

$$M_\alpha \cong \mathfrak{sl}_2(\mathbb{C})$$

Proof We check the structural constants:

$$\begin{aligned} [e_\alpha, f_\alpha] &= \left[x, \frac{2}{k(t_\alpha, t_\alpha)k(x, y)} y \right] = \frac{2}{k(t_\alpha, t_\alpha)k(x, y)} [x, y] = \frac{2}{k(t_\alpha, t_\alpha)} t_\alpha = h_\alpha \\ [e_\alpha, h_\alpha] &= \frac{2}{k(t_\alpha, t_\alpha)} [x, t_\alpha] = -\frac{2}{k(t_\alpha, t_\alpha)} \alpha(t_\alpha) x = -\frac{2}{k(t_\alpha, t_\alpha)} k(t_\alpha, t_\alpha) x = 2e_\alpha \end{aligned}$$

and

$$\begin{aligned} [f_\alpha, h_\alpha] &= \frac{4}{k(t_\alpha, t_\alpha)^2 k(x, y)} [y, t_\alpha] \\ &= -\frac{4}{k(t_\alpha, t_\alpha)^2 k(x, y)} \alpha(t_\alpha) y \\ &= -\frac{4}{k(t_\alpha, t_\alpha)^2 k(x, y)} k(t_\alpha, t_\alpha) y \\ &= -\frac{4}{k(t_\alpha, t_\alpha)k(x, y)} y \\ &= -2f_\alpha \end{aligned}$$

It has the same structural constants as $\mathfrak{sl}_2(\mathbb{C})$ so M_α is isomorphic to $\mathfrak{sl}_2(\mathbb{C})$. ■

Proposition 6.2.5

Let L be a semisimple Lie algebra over $\mathbb{C}$. Let α be a root of L . Let V be an M_α -submodule of L , then the eigenvalues of the map $\text{ad}(h_\alpha)$ are integers.

6.3 The Dimension of the Subspaces in Root Space Decompositions

Lemma 6.3.1

Let L be a semisimple Lie algebra over $\mathbb{C}$. Let Φ be the set of roots relative to some Cartan subalgebra of L . Let $\alpha \in \Phi$. Then there is a decomposition

$$H \oplus \bigoplus_{c\alpha \in \Phi} L_{c\alpha} = \ker(\alpha) \oplus M_\alpha$$

Proof Write $U_\alpha = H \oplus \bigoplus_{c\alpha \in \Phi} L_{c\alpha}$. Write $e_\alpha, f_\alpha, h_\alpha$ as the basis of M_α . Recall that M_α is spanned by $\{x, y, [x, y]\}$ where $x \in L_\alpha, y \in L_{-\alpha}$. Since $[L_\alpha, L_{-\alpha}] \subseteq L_0 = H$, we have that $M_\alpha \subseteq U_\alpha$.

Step 1: U_α is an M_α -submodule of L .

Let $h \in H$. Then we have

$$\begin{aligned} [e_\alpha, h] &= -[h, e_\alpha] = -\alpha(h)e_\alpha \in M_\alpha \subseteq U_\alpha \\ [f_\alpha, h] &= -[h, f_\alpha] = \alpha(h)f_\alpha \in M_\alpha \subseteq U_\alpha \\ [h_\alpha, h] &= 0 \end{aligned} \quad (\text{Since } H = C_L(H))$$

Hence $[M_\alpha, U_\alpha] \subseteq U_\alpha$. Now for any $c\alpha \in \Phi$, we have

$$\begin{aligned} [e_\alpha, L_{c\alpha}] &\subseteq L_{(c+1)\alpha} \subseteq U_\alpha \\ [f_\alpha, L_{c\alpha}] &\subseteq L_{(c-1)\alpha} \subseteq U_\alpha \\ [h_\alpha, L_{c\alpha}] &\subseteq L_{c\alpha} \subseteq U_\alpha \end{aligned}$$

so we are done.

Step 2: $\ker(\alpha)$ is a trivial M_α -submodule of U_α .

Let $h \in \ker(\alpha) \subseteq H$. Then we compute that

$$\begin{aligned} [e_\alpha, h] &= -[h, e_\alpha] = -\alpha(h)e_\alpha = 0 \\ [f_\alpha, h] &= -[h, f_\alpha] = \alpha(h)f_\alpha = 0 \\ [h_\alpha, h] &= 0 \end{aligned} \quad (\text{Since } H = C_L(H))$$

So we are done.

Step 3: M_α is an irreducible M_α -submodule of U_α .

Clearly M_α is a submodule of U_α . Since sub-modules of M_α correspond bijectively to ideals of M_α and $M_\alpha \cong \mathfrak{sl}_2(\mathbb{C})$ is simple, we conclude that M_α is irreducible.

Step 4: $\ker(\alpha) \oplus M_\alpha$ is an M_α -submodule of U_α .

It suffices to show that $\ker(\alpha) \cap M_\alpha = \{0\}$. Since $\ker(\alpha) \cap M_\alpha \subseteq H$ and $H \cap M_\alpha = \mathbb{C}\langle h_\alpha \rangle$, as well as $e_\alpha, f_\alpha \notin L_0 = H$, we conclude that $\ker(\alpha) \cap M_\alpha = \{0\}$.

By Weyl's theorem, there exists an M_α -submodule of U_α so that

$$U_\alpha = M_\alpha \oplus \ker(\alpha) \oplus W$$

Step 5: $W = \{0\}$.

Suppose that W is non-zero. Then there exists an irreducible M_α -submodule V of W . By the classification of irreducible $\mathfrak{sl}_2(\mathbb{C})$ -modules, we know that $V \cong W_d$ for some $d \in \mathbb{N}$. By 6.4.1 and 6.4.2, we know that the eigenvalues of $\text{ad}(h_\alpha)$ is given by $d, d-2, \dots, -d$. They are either all even or all odd.

If d is even, then $\text{ad}(h_\alpha)$ has an eigenvector $0 \neq v \in V$ whose eigenvalue is 0. Now the eigenspace of $\text{ad}(h)$ corresponding to the eigenvalue 0 is H because $H = L_0 = C_L(H)$. Hence $v \in H$ so that $v \in H \cap V$. At the same time, $H \subseteq \ker(\alpha) \oplus M_\alpha$ and $(\ker(\alpha) \oplus M_\alpha) \cap W = \{0\}$ implies that $H \cap V = \{0\}$, a contradiction.

If d is odd, then $\text{ad}(h_\alpha)$ has an eigenvector $0 \neq v \in V$ whose eigenvalue is 1. By 9.2.4, $\alpha(h_\alpha) = 2$. The linear map $\text{ad}(h) : L_{c\alpha} \subseteq U_\alpha L_{c\alpha} \subseteq U_\alpha$ by definition satisfies the equation $\text{ad}(h_\alpha)(w) = c\alpha(h_\alpha)w = 2cw$. When $v = w$, its eigenvalue is 1 and this implies that $c = 1/2$ so that both $\alpha/2$ and α are roots. I claim that this is a contradiction. By 9.2.4, $\frac{\alpha}{2}(h_{\alpha/2}) = 2$ so that $\text{ad}(\alpha/2)$ has eigenvalue 2 on $L_\alpha \subseteq U_{\alpha/2}$. Then the eigenvalues of $\text{ad}(h_{\alpha/2})$ on $\ker(\alpha/2) \oplus M_{\alpha/2}$ is 0 and ± 2 . This implies that W_d is a submodule of W for some even d , and by the even case this is impossible. Hence we reached a contradiction.

The fact that $W = \{0\}$ then implies that $U_\alpha = \ker(\alpha) \oplus M_\alpha$ as required. ■

Proposition 6.3.2

Let L be a semi-simple Lie algebra over $\mathbb{C}$. Let Φ be the set of roots relative to some Cartan sub-algebra of L . Let $\alpha \in \Phi$. If $c\alpha \in \Phi$, then $c = \pm 1$.

Proof We know that $H \oplus \bigoplus_{c\alpha} L_{c\alpha} = \ker(\alpha) \oplus M_\alpha$. Analyzing the right hand side, the fact that $\alpha(h_\alpha) = 2$ implies that the only eigenvalues for $\text{ad}(h_\alpha)$ on $\ker(\alpha) \oplus M_\alpha$ can only be 0 or ± 2 . Analyzing the left hand side, the eigenvalue of $\text{ad}(h_\alpha)$ on H is 0 and the eigenvalue of $\text{ad}(h)$ on $L_{c\alpha}$ is $c\alpha$ by construction. This proves that $c = \pm 1$. ■

Proposition 6.3.3

Let L be a semi-simple Lie algebra over $\mathbb{C}$. Let Φ be the set of roots relative to some Cartan subalgebra of L . Let $\alpha \in \Phi$. Then we have

$$\dim(L_\alpha) = 1$$

Proof We have that

$$\dim(H) + 2 = \dim(H \oplus L_\alpha \oplus L_{-\alpha}) = \dim(\ker(\alpha) \oplus M_\alpha) = \dim(\ker \alpha) + 3$$

$\dim(\ker(\alpha)) = 1$ since $\alpha : H \rightarrow \mathbb{C}$ is a root. Since both L_α and $L_{-\alpha}$ are non-trivial vector spaces, we are left with $\dim(L_\alpha) + \dim(L_{-\alpha}) = 2$ which implies that $\dim(L_\alpha) = 1$. ■

6.4 Root Strings

Definition 6.4.1 (Root Strings)

Let L be a semisimple Lie algebra over $\mathbb{C}$. Let Φ be a root system of L . Let $\alpha \in \Phi$ and $\beta \in \Phi \cup \{0\}$. Define the α -root string through β to be

$$S_{\alpha,\beta} = \bigoplus_{\substack{c \in \mathbb{Z} \\ \beta + c\alpha \in \Phi}} L_{\beta + c\alpha}$$

Lemma 6.4.2

Let L be a semisimple Lie algebra over $\mathbb{C}$. Let Φ be a root system of L . Let $\alpha, \beta \in \Phi$ such that $\beta \neq \pm\alpha$. Then the following are true.

- There exists $k_1, k_2 \in \mathbb{Z}$ such that for any $c \in \mathbb{Z}$, $\beta + c\alpha \in \Phi$ if and only if $k_2 < c < k_1$. Moreover, $\beta(h_\alpha) = k_1 - k_2$.
- $\beta - \beta(h_\alpha)\alpha \in \Phi$.

Proposition 6.4.3

Let L be a semisimple Lie algebra over $\mathbb{C}$. Let Φ be a root system of L . Let $\alpha \in \Phi$ and $\beta \in \Phi \cup \{0\}$. Then $S_{\alpha,\beta}$ is an irreducible $M_\alpha \cong \mathfrak{sl}_2(\mathbb{C})$ -module.

Example 6.4.4

Let $\varepsilon_i \in H^*$ be the map defined by $\varepsilon_i(e_{j,j}) = \delta_{i,j}$. Recall that a root system of $\mathfrak{sl}_3(\mathbb{C})$ is given by $\Phi = \{\varepsilon_i - \varepsilon_j \mid 1 \leq i, j \leq 3\}$ and a base for the root system is given by

$$\{\alpha_1 = \varepsilon_1 - \varepsilon_2, \alpha_2 = \varepsilon_2 - \varepsilon_3\}$$

- The α_2 -root string through α_1 is given by

$$L_{\alpha_1} \oplus L_{\alpha_1 + \alpha_2} = \mathbb{C}\langle e_{1,2}, e_{1,3} \rangle$$

- The $-\alpha_1$ -root string through $-\alpha_2$ is given by

$$L_{-\alpha_2} \oplus L_{-\alpha_2 - \alpha_1} = \mathbb{C}\langle e_{3,2}, e_{3,1} \rangle$$

- The α_1 -root string through 0 is given by

$$L_{-\alpha_1} \oplus L_0 \oplus L_{\alpha_1} = \mathbb{C}\langle e_{1,1} - e_{2,2}, e_{1,2}, e_{2,1} \rangle$$

6.5 The Set of Roots as a Root System

Proposition 6.5.1

Let L be a semi-simple Lie algebra over $\mathbb{C}$. Let H be a Cartan subalgebra of L . Let Φ be the set of roots relative to H . Let $\alpha_1, \dots, \alpha_r \in \Phi$ be a basis for H^* . Then $\mathbb{R}[\alpha_1, \dots, \alpha_r] \subset H^*$ is an inner product space whose symmetric bilinear form is given by

$$B(\phi, \psi) = k(t_\phi, t_\psi)$$

(as in the notation of 9.2.1).

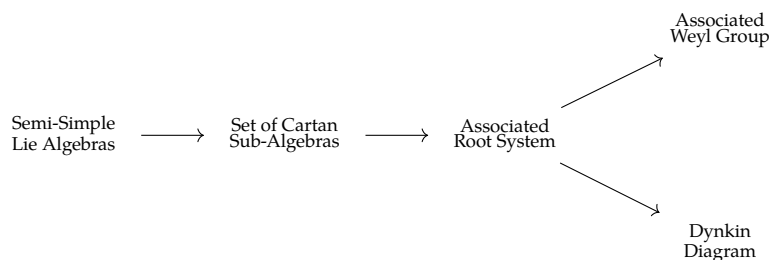
Proposition 6.5.2

Let L be a semi-simple Lie algebra over $\mathbb{C}$. Let H be a Cartan subalgebra of L . Let Φ be the set of roots relative to H . Let $\alpha_1, \dots, \alpha_r \in \Phi$ be a basis for H^* . Then Φ is a root system for the inner product space $(\mathbb{R}[\alpha_1, \dots, \alpha_r], B)$.

Proposition 6.5.3

Let L be a semi-simple Lie algebra over $\mathbb{C}$. Let H and K be two Cartan subalgebras of L . Let Φ_H and Φ_K be the set of roots relative to H and K respectively. Then $\Phi_H \cong \Phi_K$.

This proves that although the set of roots come from a chosen Cartan subalgebra of the Lie algebra L , it is an invariant of L and not the Cartan subalgebra. Up until this point, we have constructed an invariant for Lie algebras following the below schematic:



We will now show that Simple Lie algebras give rise to irreducible Dynkin diagrams and vice versa.

Example 6.5.4

The following are true regarding $\mathfrak{sl}_n(\mathbb{C})$.

- A Cartan sub-algebra of $\mathfrak{sl}_n(\mathbb{C})$ is given by $H = \mathfrak{d}_n(\mathbb{C}) \cap \mathfrak{sl}_n(\mathbb{C})$
- Let $\varepsilon_i \in H^*$ be the map defined by $\varepsilon_i(e_{j,j}) = \delta_{i,j}$. Then the set of roots of L with respect to H is given by

$$\Phi = \{\varepsilon_i - \varepsilon_j \mid 1 \leq i, j \leq n\}$$

- A copy of $\mathfrak{sl}_2(\mathbb{C})$ in $\mathfrak{sl}_n(\mathbb{C})$ is given by

$$\mathbb{C}\langle e_{i,j}, e_{j,i}, e_{i,i} - e_{j,j} \rangle$$

for $i \neq j$.

- A base for Φ is given by

$$\{\varepsilon_i - \varepsilon_{i+1} \mid 1 \leq i \leq n-1\}$$

6.6 The Classification Theorem

Proposition 6.6.1

Let L be a semisimple Lie algebra over $\mathbb{C}$. Let Φ be a root system of L with base $B = \{\alpha_1, \dots, \alpha_r\}$.

Let $\mathfrak{sl}_2(\mathbb{C}) \cong M_{\alpha_i}$ be the corresponding Lie subalgebra of L generated by the elements

$$e_i = \Phi \left(\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \right) \quad f_i = \Phi \left(\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \right) \quad h_i = \Phi \left(\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right)$$

Then the following are true.

- $L = \mathbb{C}\langle e_i, f_i \mid 1 \leq i \leq r \rangle$.
- $[h_s, h_t] = 0$ for $1 \leq s, t \leq l$.
- $[h_s, e_t] = \langle \alpha_t, \alpha_s \rangle e_t$ for $1 \leq s, t \leq l$.
- $[h_s, f_t] = -\langle \alpha_t, \alpha_s \rangle f_t$ for $1 \leq s, t \leq l$.
- $[e_s, f_t] = \begin{cases} h_s & \text{if } s = t \\ 0 & \text{otherwise} \end{cases}$
- $(\text{ad}(e_s))^{1-\langle \alpha_t, \alpha_s \rangle} (e_t) = 0 = (\text{ad}(f_s))^{1-\langle \alpha_t, \alpha_s \rangle} (f_t)$ for $1 \leq s, t \leq l$.

Theorem 6.6.2 (Serre's Theorem)

Let R be a root system over $\mathbb{R}^n$ with base $B = \{\alpha_1, \dots, \alpha_l\}$. Let L be the Lie algebra defined by

$$L = \mathbb{C}\langle e_i, f_i, h_i, 1 \leq i \leq l \mid S \rangle$$

where S are the following relations:

- $[h_s, h_t] = 0$ for $1 \leq s, t \leq l$.
- $[h_s, e_t] = \langle \alpha_t, \alpha_s \rangle e_t$ for $1 \leq s, t \leq l$.
- $[h_s, f_t] = -\langle \alpha_t, \alpha_s \rangle f_t$ for $1 \leq s, t \leq l$.
- $[e_s, f_t] = \begin{cases} h_s & \text{if } s = t \\ 0 & \text{otherwise} \end{cases}$
- $(\text{ad}(e_s))^{1-\langle \alpha_t, \alpha_s \rangle} (e_t) = 0 = (\text{ad}(f_s))^{1-\langle \alpha_t, \alpha_s \rangle} (f_t)$ for $1 \leq s, t \leq l$.

Then the following are true.

- L is finite dimensional and semi-simple.
- $H = \mathbb{C}\langle h_1, \dots, h_l \rangle$ is a Cartan sub-algebra of L .
- The root system Φ_H of H is equal to R .

Corollary 6.6.3

There is a bijection

$$\frac{\text{Semisimple Lie algebras over } \mathbb{C}}{\cong} \xleftrightarrow{1:1} \frac{\text{Root Systems over } \mathbb{R}^n \text{ for all } n}{\cong}$$

given as follows.

- For a semisimple Lie algebra over $\mathbb{C}$, choose a Cartan sub-algebra H of L . Then H has an associated root system.
- For a root system R , Serre's theorem gives a semisimple Lie algebra over $\mathbb{C}$.

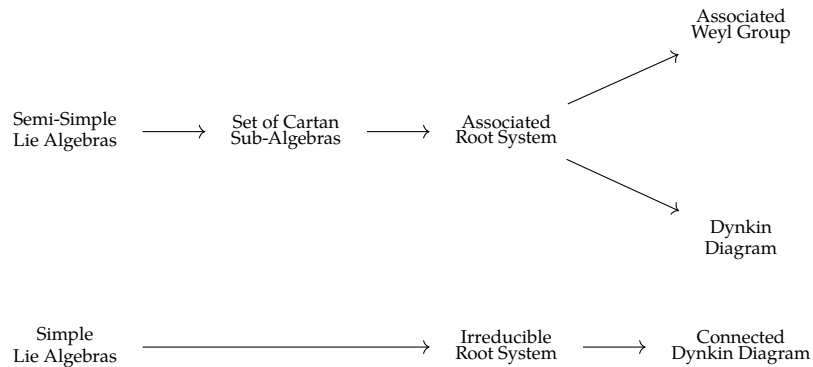
Proof The forward map is well defined since the associated root system is independent of the chosen Cartan sub-algebra by 9.3.4. The backwards map is well defined by construction.

Let L be a semisimple Lie algebra over $\mathbb{C}$. Then using its root system, we can construct a new semisimple Lie algebra over $\mathbb{C}$ by Serre's theorem. We need to show that these two semisimple Lie algebras over $\mathbb{C}$ are isomorphic.

Given a root system R , Serre's theorem gives a semisimple Lie algebra over $\mathbb{C}$ whose associated root system is equal to R . Hence the backward maps composed with the forward map is the identity. ■

6.7 Simple Lie Algebras and Irreducible Dynkin Diagrams

We now want to enrich the above diagram:



Proposition 6.7.1

Let L be a simple Lie algebra over $\mathbb{C}$. Let H be a Cartan sub-algebra of L . Let Φ be the root system associated to H . Then Φ is irreducible.

Corollary 6.7.2

Let Φ be an irreducible root system. Then there exists a simple Lie algebra over $\mathbb{C}$ such that its associated root system is Φ .

Root System	Rank	Simple Lie Algebra	Dimension	Number of Roots
A_n ($n \geq 1$)	n	$\mathfrak{sl}_{n+1}(\mathbb{C})$	$n^2 + 2n$	$n^2 + n$
B_n ($n \geq 2$)	n	$\mathfrak{so}_{2n+1}(\mathbb{C})$	$2n^2 + n$	$2n^2$
C_n ($n \geq 3$)	n	$\mathfrak{sp}_{2n}(\mathbb{C})$	$2n^2 + n$	$2n^2$
D_n ($n \geq 4$)	n	$\mathfrak{so}_{2n}(\mathbb{C})$	$2n^2 - n$	$2n^2 - 2n$
G_2	2	$\mathfrak{g}_2(\mathbb{C})$	14	12
F_4	4	$\mathfrak{f}_4(\mathbb{C})$	52	48
E_6	6	$\mathfrak{e}_6(\mathbb{C})$	78	72
E_7	7	$\mathfrak{e}_7(\mathbb{C})$	133	126
E_8	8	$\mathfrak{e}_8(\mathbb{C})$	248	240

Corollary 6.7.3

There are no semisimple Lie algebras of dimension 1, 2, 4, 5, 7.

Another table of simple Lie algebras, indexed by the dimension.

Dimension	Simple Lie Algebras	Root System
3	$\mathfrak{sl}_2(\mathbb{C})$	A_1
8	$\mathfrak{sl}_3(\mathbb{C})$	A_2
10	$\mathfrak{so}_5(\mathbb{C})$	B_2
14	$\mathfrak{g}_2(\mathbb{C})$	G_2
15	$\mathfrak{sl}_4(\mathbb{C})$	A_3
21	$\mathfrak{so}_7(\mathbb{C}), \mathfrak{sp}_6(\mathbb{C})$	B_3, C_3
24	$\mathfrak{sl}_5(\mathbb{C})$	A_4
28	$\mathfrak{so}_8(\mathbb{C})$	D_4

Finally, a table of semisimple Lie algebras.

Dimension	Lie Algebras	Total Number
3	$\mathfrak{sl}_2(\mathbb{C})$	1
6	$\mathfrak{sl}_2(\mathbb{C})^{\oplus 2}$	1
8	$\mathfrak{sl}_3(\mathbb{C})$	1
9	$\mathfrak{sl}_2(\mathbb{C})^{\oplus 3}$	1
10	$\mathfrak{so}_5(\mathbb{C})$	1
11	$\mathfrak{sl}_2(\mathbb{C}) \oplus \mathfrak{sl}_3(\mathbb{C})$	1
12	$\mathfrak{sl}_2(\mathbb{C})^{\oplus 4}$	1
13	$\mathfrak{sl}_2(\mathbb{C}) \oplus \mathfrak{so}_5(\mathbb{C})$	1
14	$\mathfrak{g}_2(\mathbb{C}), \mathfrak{sl}_2(\mathbb{C})^{\oplus 2} \oplus \mathfrak{sl}_3(\mathbb{C})$	2
15	$\mathfrak{sl}_4(\mathbb{C}), \mathfrak{sl}_2(\mathbb{C})^{\oplus 5}$	2
16	$\mathfrak{sl}_3(\mathbb{C})^{\oplus 2}, \mathfrak{sl}_2(\mathbb{C})^{\oplus 2} \oplus \mathfrak{so}_5(\mathbb{C})$	2
17	$\mathfrak{sl}_2(\mathbb{C})^{\oplus 3} \oplus \mathfrak{sl}_3(\mathbb{C}), \mathfrak{sl}_2(\mathbb{C}) \oplus \mathfrak{g}_2(\mathbb{C})$	2
18	$\mathfrak{sl}_2(\mathbb{C})^{\oplus 6}, \mathfrak{sl}_3(\mathbb{C}) \oplus \mathfrak{so}_5(\mathbb{C}), \mathfrak{sl}_4(\mathbb{C}) \oplus \mathfrak{sl}_2(\mathbb{C})$	3
19	$\mathfrak{sl}_2(\mathbb{C}) \oplus \mathfrak{sl}_3(\mathbb{C})^{\oplus 2}, \mathfrak{sl}_2(\mathbb{C})^{\oplus 3} \oplus \mathfrak{so}_5(\mathbb{C})$	2
20	$\mathfrak{sl}_2(\mathbb{C})^{\oplus 4} \oplus \mathfrak{sl}_3(\mathbb{C}), \mathfrak{sl}_2(\mathbb{C})^{\oplus 2} \oplus \mathfrak{g}_2(\mathbb{C}), \mathfrak{so}_5(\mathbb{C})^{\oplus 2}$	3